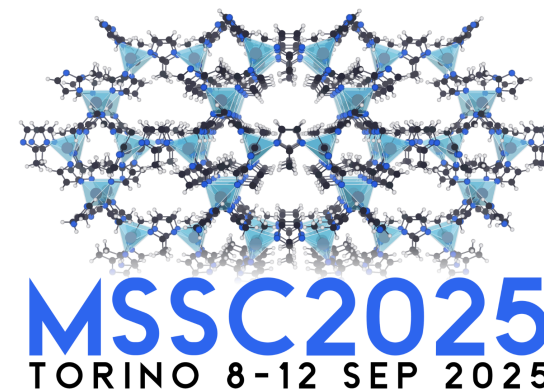


Modelling IR and Raman Spectra and Non-Linear Optical Properties

Lorenzo Maschio

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lorenzo.maschio@unito.it



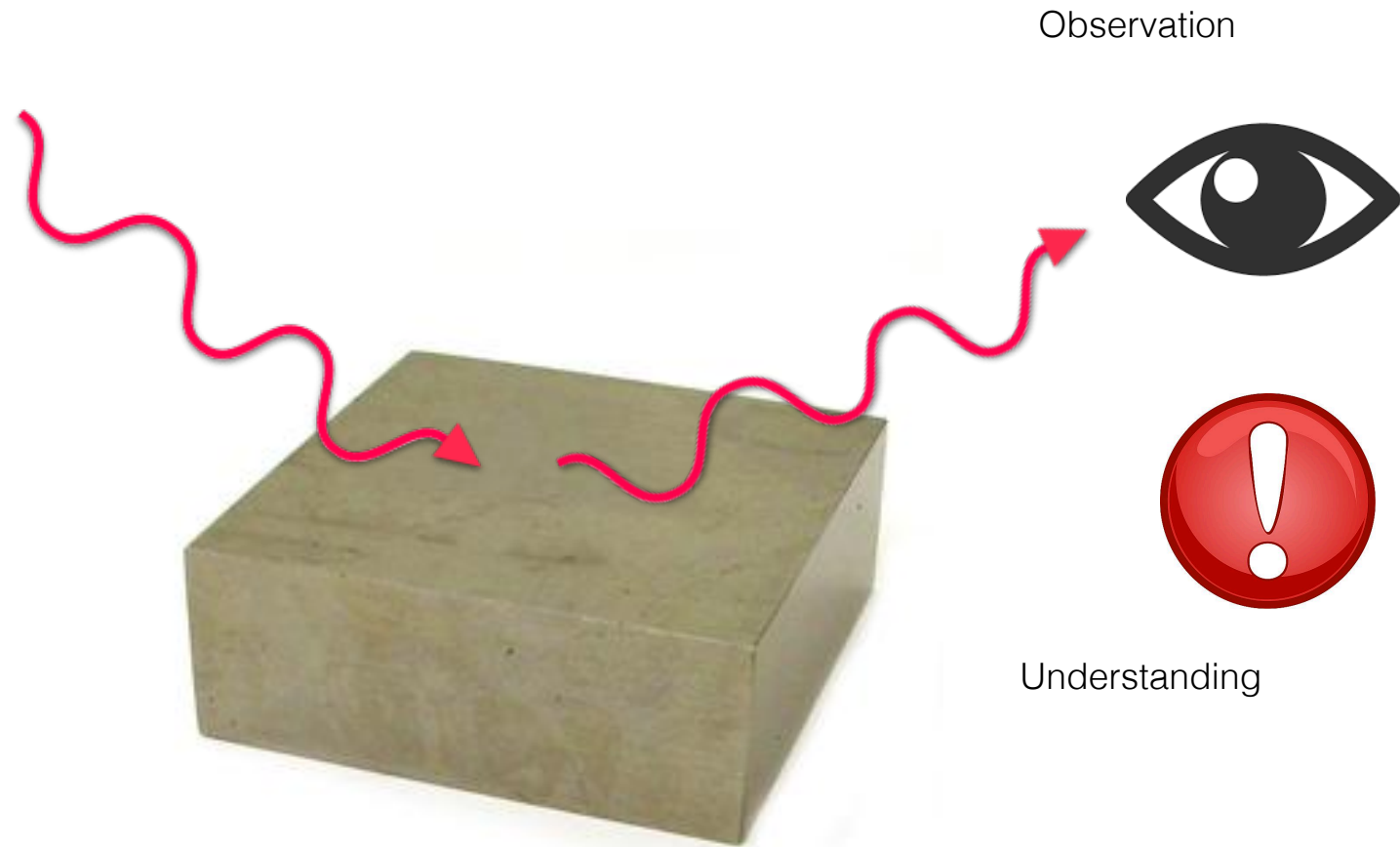
How do we investigate matter?



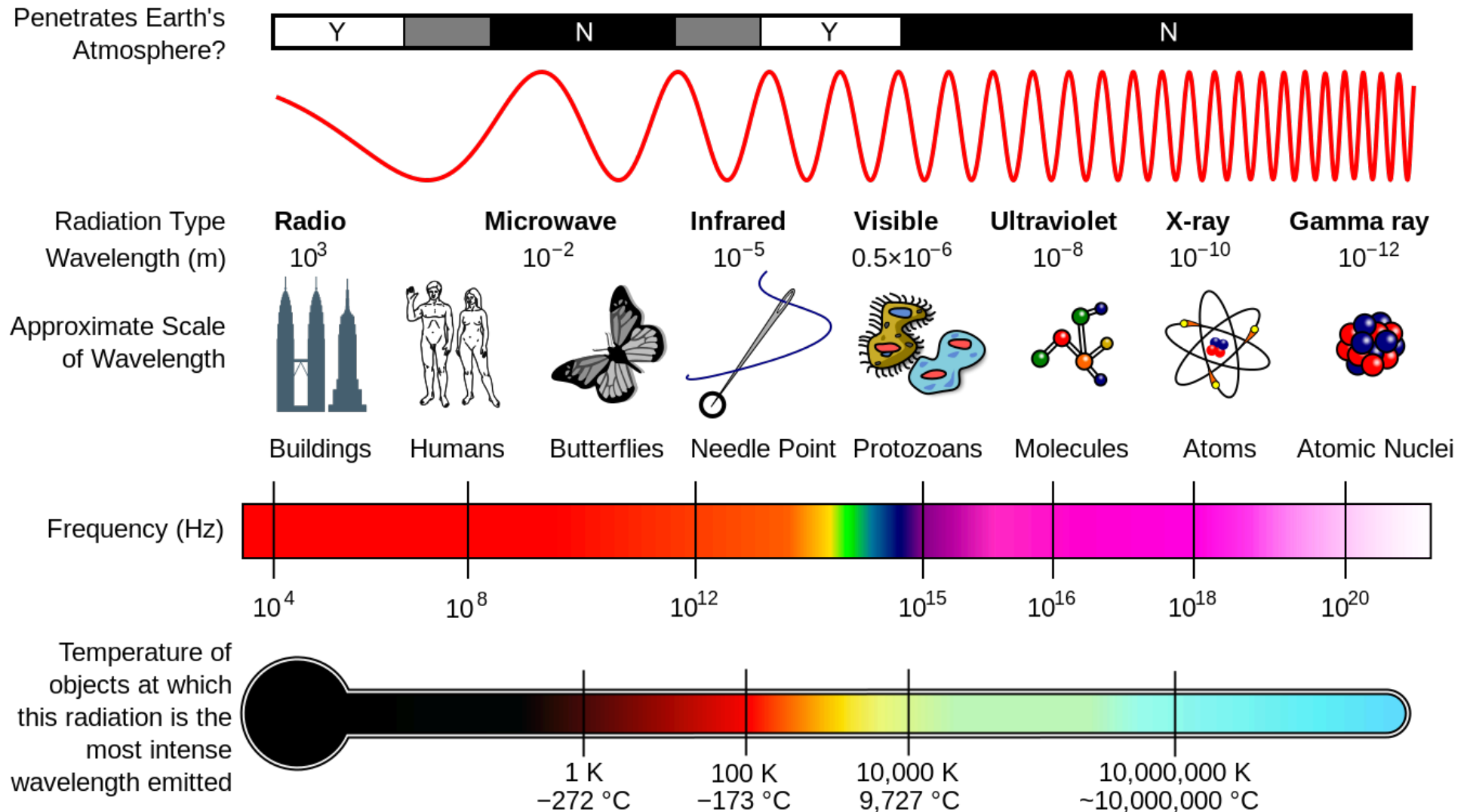
How do we investigate matter?



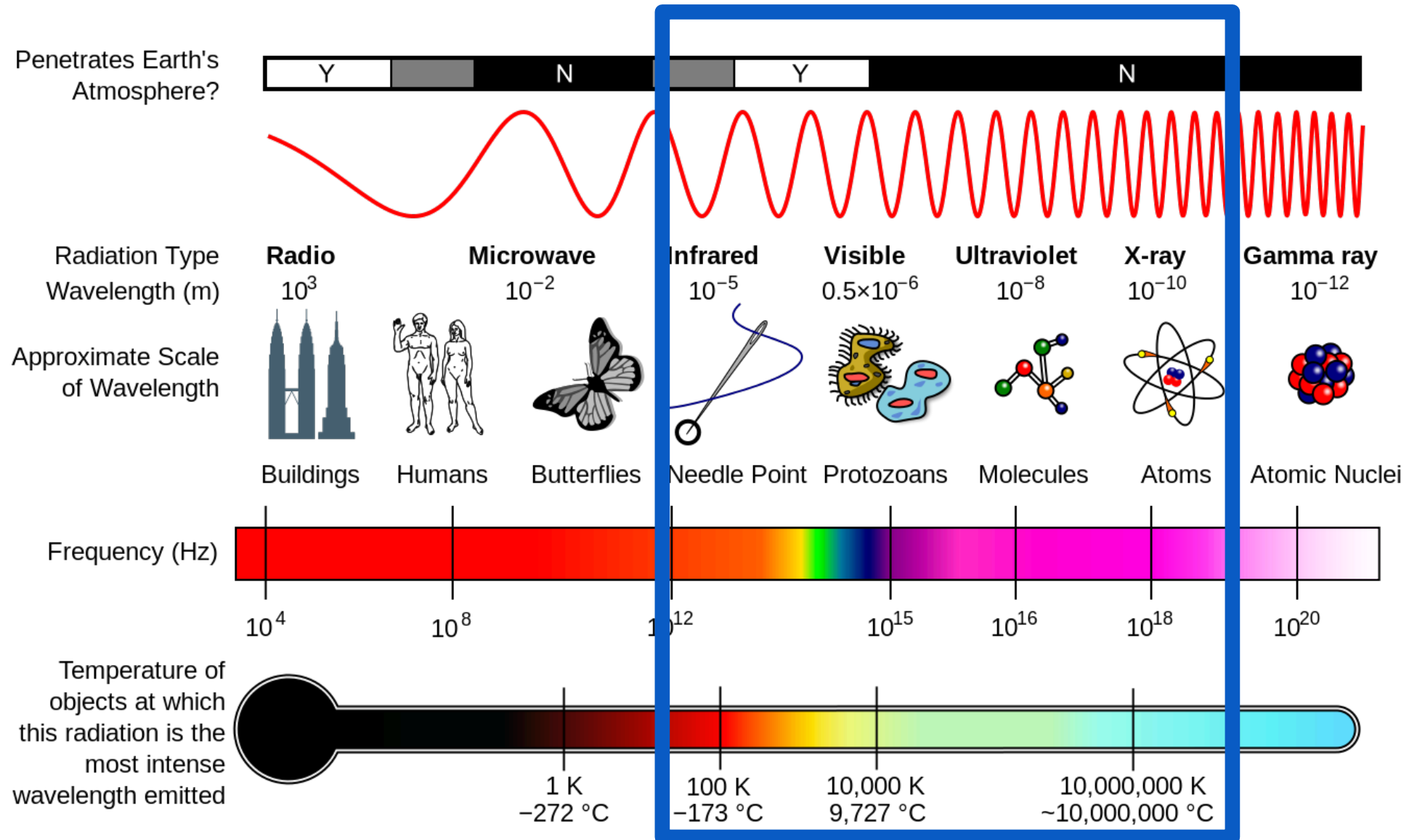
Light is very useful to investigate matter



Which light?

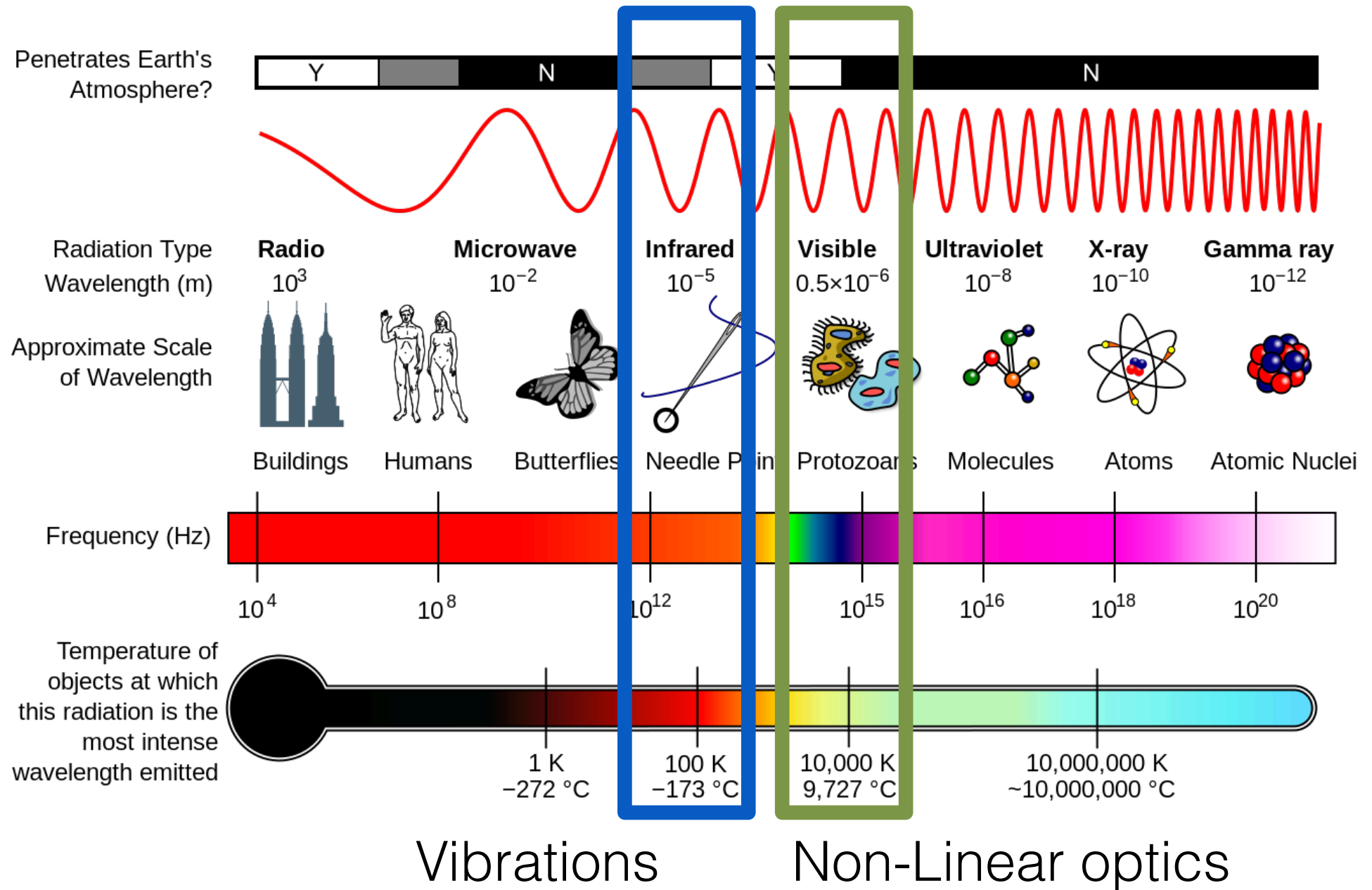


Which light?

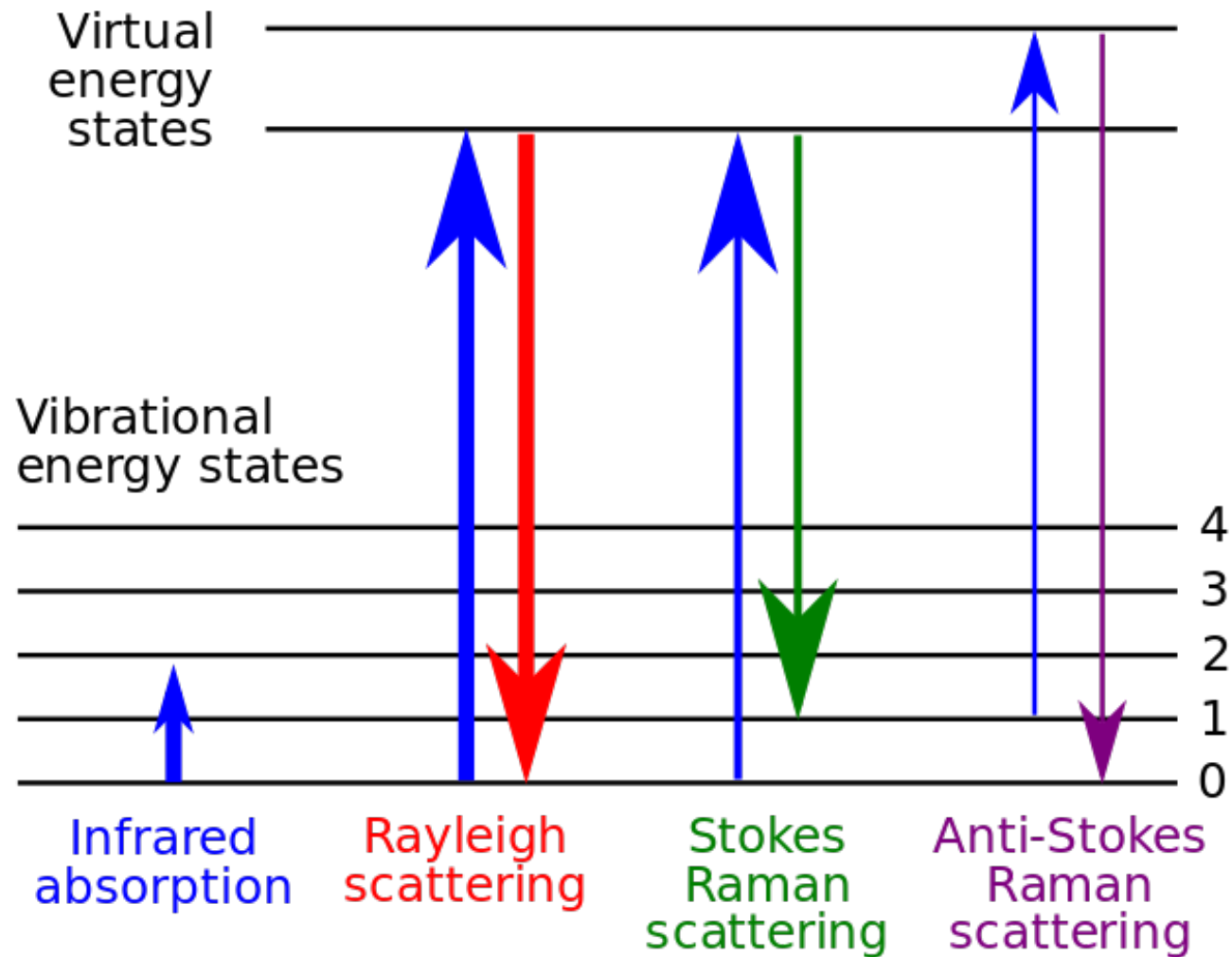


Interaction with matter

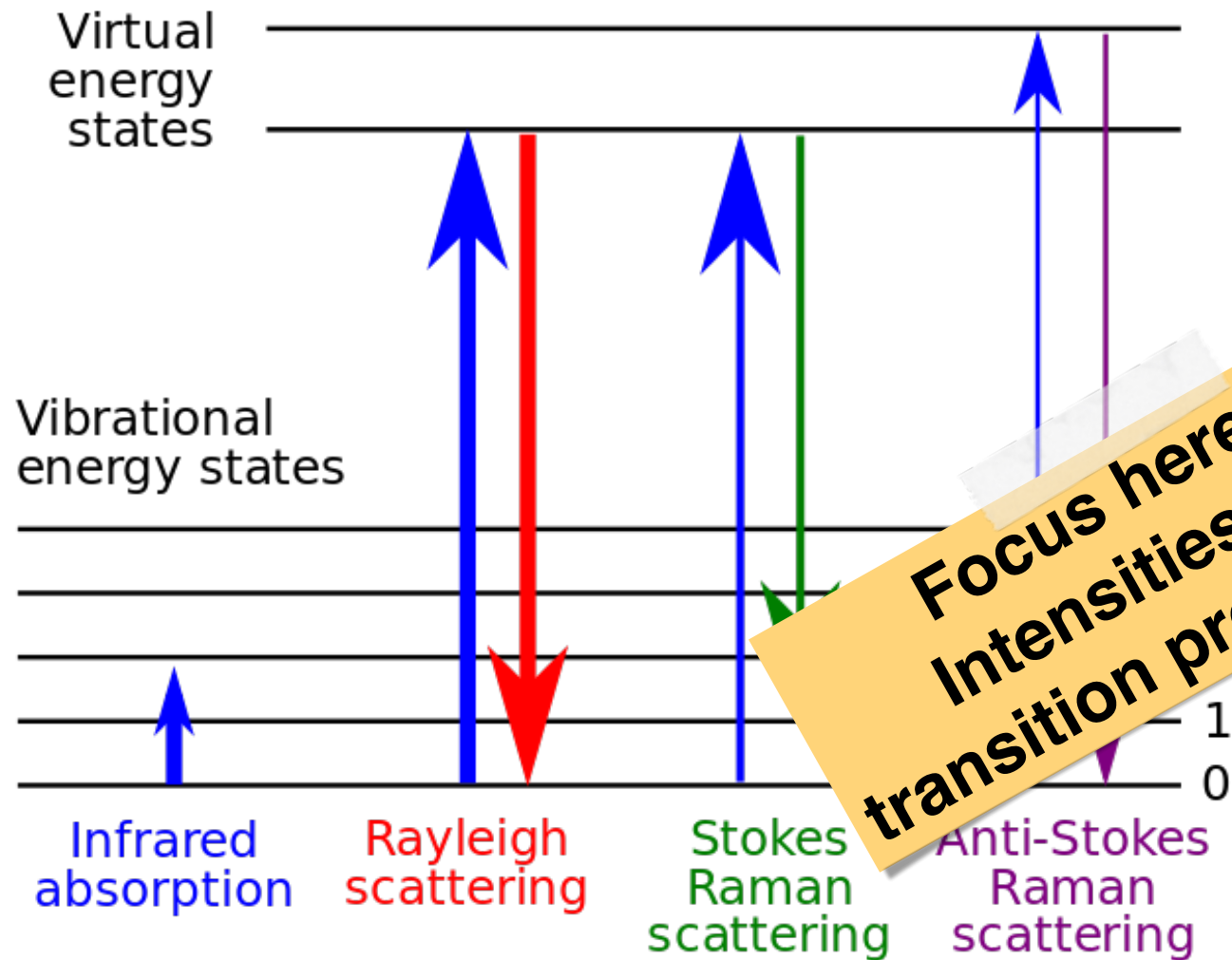
Which light?



Infrared and Raman absorption

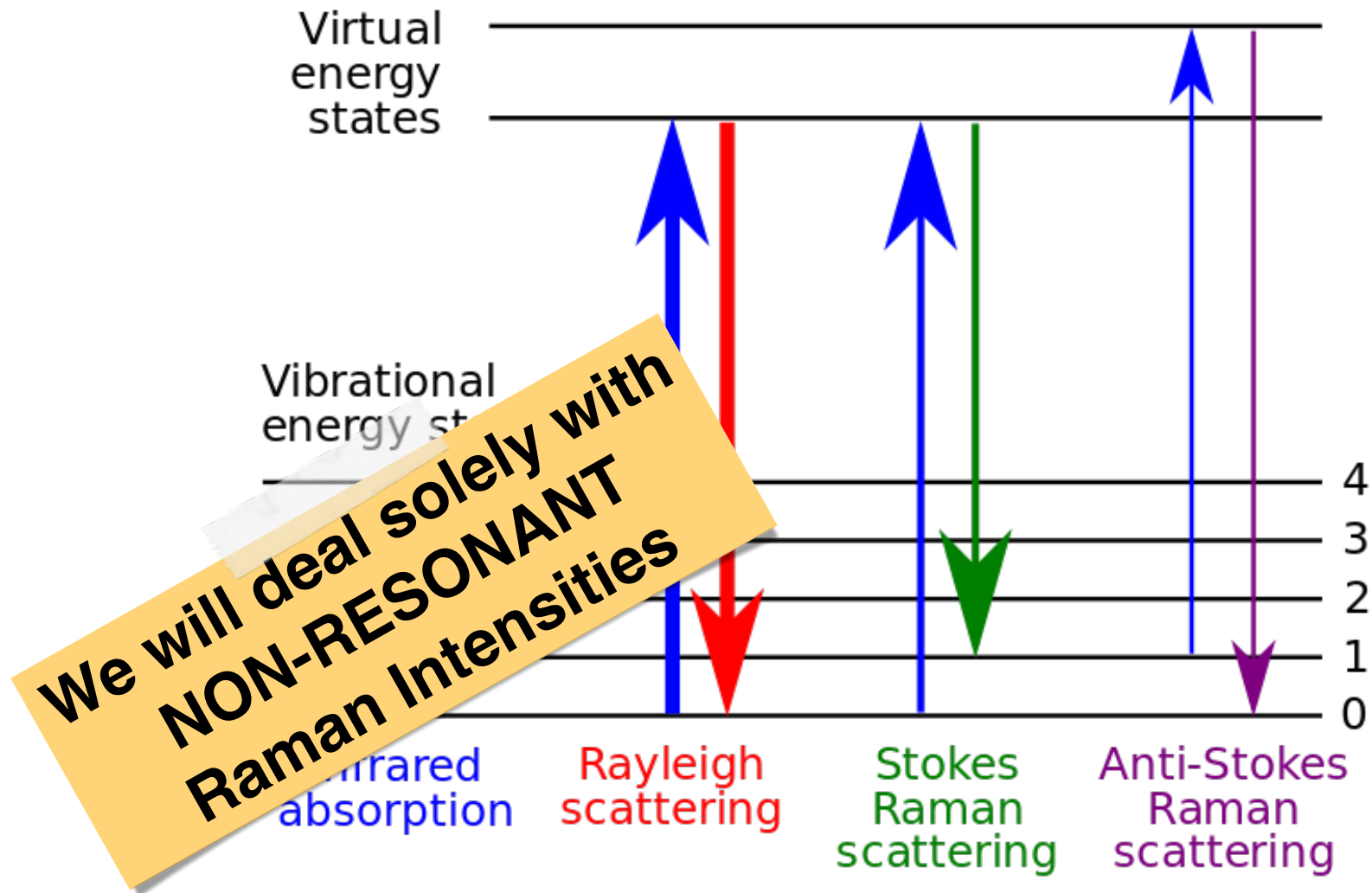


Infrared and Raman absorption



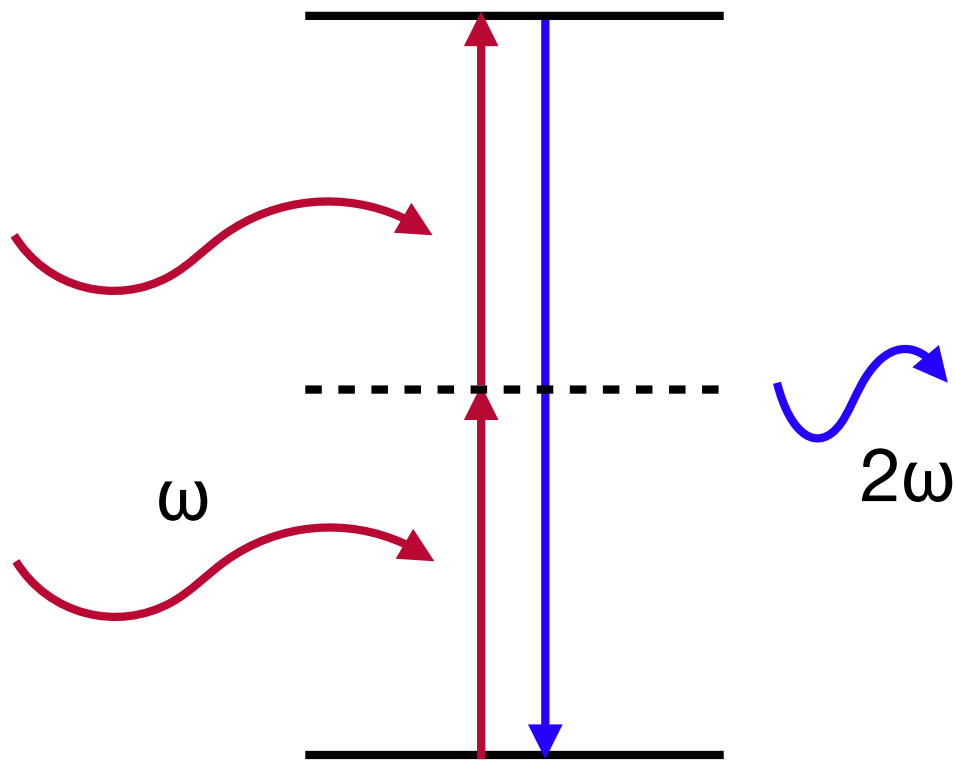
**Focus here on
Intensities, (i.e.
transition probabilities)**

Infrared and Raman absorption

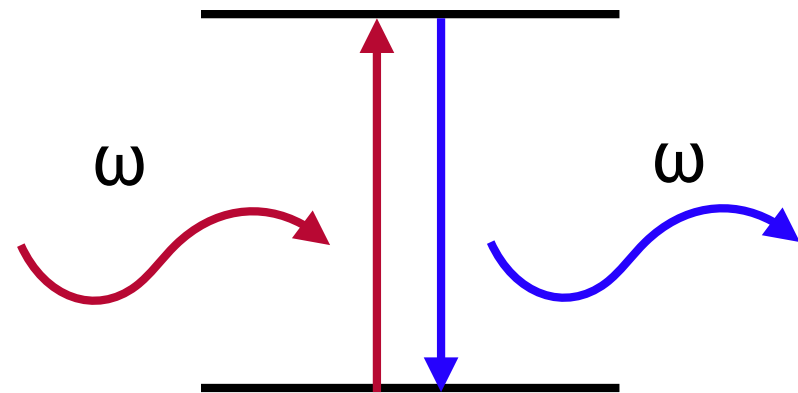


SHG and Pockels

Second Harmonic Generation



Pockels Effect



Today's menu

Today's menu

Appetizer



Today's menu

Appetizer

Main course - Theory



Today's menu

Appetizer

Main course - Theory

Cheese - From theory to experiment



Today's menu

Appetizer

Main course - Theory

Cheese - From theory to experiment

Dessert - Some simulated spectra



Today's menu

Appetizer

Main course - Theory

Cheese - From theory to experiment

Dessert - Some simulated spectra

Coffee



1. The Appetizer

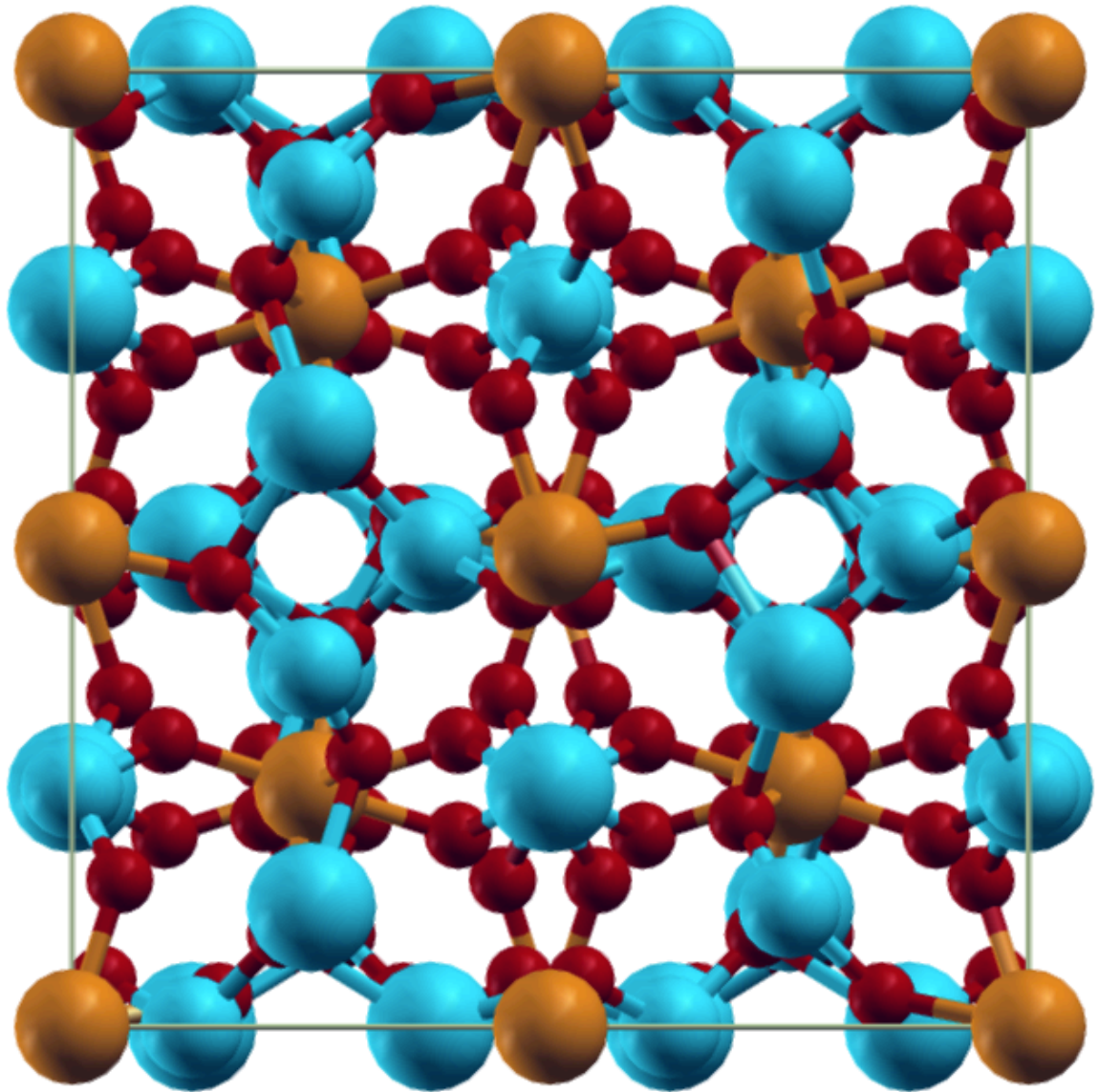


Pyrope

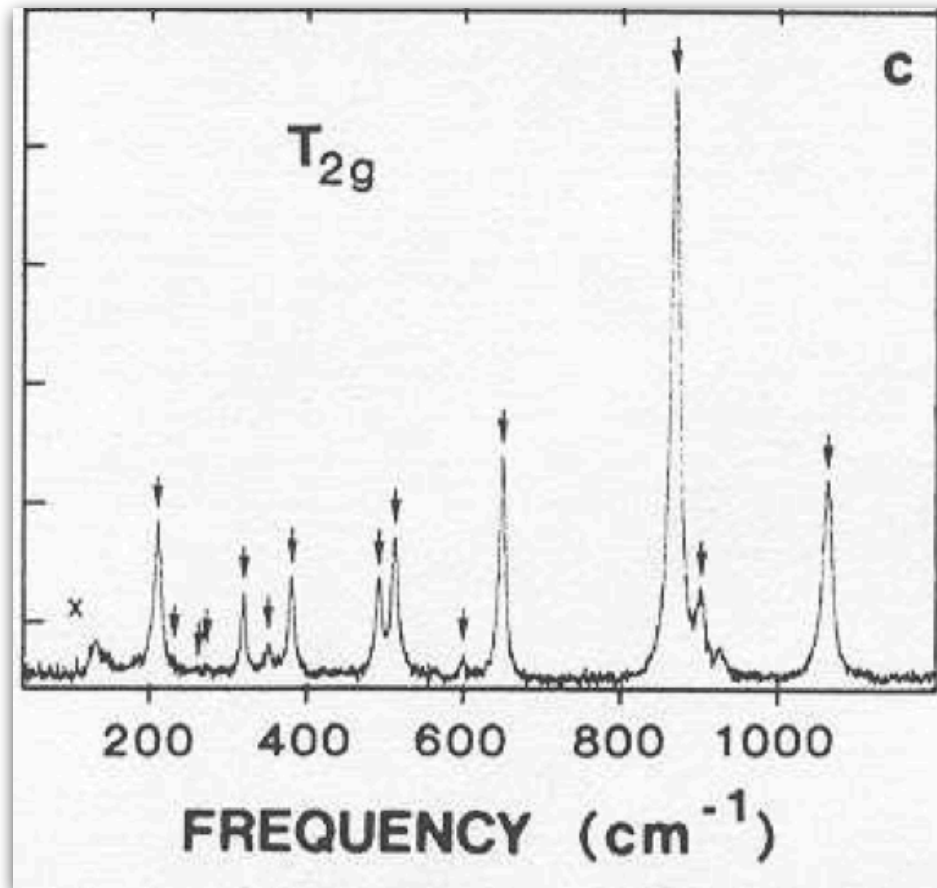


Cubic,
80 atoms in the
unit cell

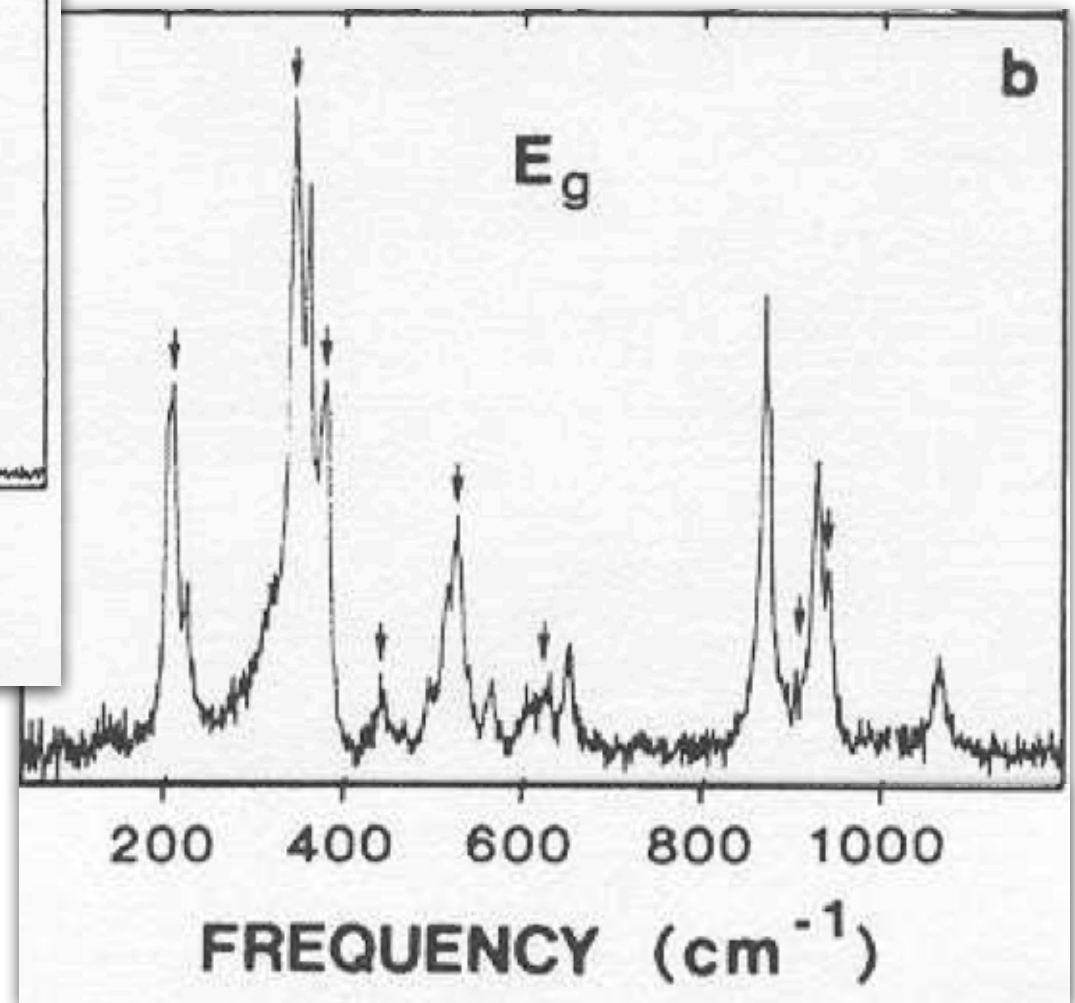
Raman
spectrum,
a long story



Pyrope



Hofmeister et al. 1991



Pyrope

Hofmeister et al. 1991

All 25 Raman active modes were assigned

Number *	Symmetry	Assignment	Pyrope		
			Raman	LO	TO ^a
1B	$T_{2g} + T_{1u}$	ν_3	1062	1063	976
2	E_g	ν_3	938		
3C	$T_{2g} + T_{1u}$	ν_3	899	889	906
4D	$T_{2g} + T_{1u}$	ν_3	866	940	878
5	A_{1g}	ν_1	925		
6	E_g	ν_1	911		
7E	$T_{2g} + T_{1u}$	ν_4	648	667	664
8	E_g	ν_4	626		
9F	$T_{2g} + T_{1u}$	ν_4	598	618	583
10G	$T_{2g} + T_{1u}$	ν_4	510	528	536
11	A_{1g}	ν_2	562		
12	E_g	ν_2	524		
13I	$T_{2g} + T_{1u}$	ν_2	490	556	458
14	E_g	ν_2	439		
15K	$T_{2g} + T_{1u}$	R(SiO ₄)	379	400	385
16	E_g	R(SiO ₄)	365		
17	A_{1g}	R(SiO ₄)	362		
18L	$T_{2g} + T_{1u}$	R(SiO ₄)	350	370	365
19H	T_{1u}	A(Al)	—	474	478
20J	T_{1u}	T(Al)	—	422	423
21N	T_{1u}	T(Al)	—	280	279
22O	$T_{2g} + T_{1u}$	T(M)	272 ^c	263	260
23P	T_{1u}	T(M)		240	238
24Q	$T_{2g} + T_{1u}$	T(M)	208	218	200
25	E_g	T(M)	203		
26	E_g	T(SiO ₄)	342		
27M	$T_{2g} + T_{1u}$	T(SiO ₄)	318	353	339
28	T_{2g}	T(SiO ₄) mix	285 ^c		
29R	$T_{2g} + T_{1u}$	T(SiO ₄) mix	230	152	140

Pyrope

Sym.	Hofmeister - Chopelas 1991				
Eg	-				
	342				
	365				
	439				
	911				
F2g	-				
	-				
	230				
	285				
	318				

Experiment

Simulation

Pyrope



Calculated modes and assignments		
Mode†	Mode‡	Assignment‡
925	850.5	ν_1
562	524.0	ν_2
362	342.7	R(SiO ₄)
938	943.1	ν_3
911(867§)	816.3	ν_1
626	633.2	ν_4
524	506.8	ν_2
439	430.6	ν_2
365(379§)	363.8	R(SiO ₄)
342(309§)	307.7	R(SiO ₄)
		T(Mg)
203	207.2	T(SiO ₄)
		T(Mg)
1062	1061.8	ν_3
899	875.8	ν_3
866	844.4	ν_3
648	643.8	ν_4
598	607.2	ν_4
510	514.9	ν_4
490	473.2	ν_2
379	367.3	R(SiO ₄)
350	353.1	R(SiO ₄)
		T(Mg)
318(342§)	323.1	T(SiO ₄)
		?R(SiO ₄)
285(318§)	297.2	T(Mg)
		T(SiO ₄)
272	246.6	T(Mg)
		T(SiO ₄)
230	227.4	T(Mg)
208	192.6	T(Mg)

Chaplin et al. 1998

Method:
Classical dynamics

Pyrope

Sym.	Hofmeister - Chopelas 1991		Chaplin et al. 1998		
Eg	-		308		
	342				
	365		364		
	439		431		
	911		816		
F2g	-		-		
	-		-		
	230				
	285		297		
	318		323		

Experiment

Simulation

Pyrope

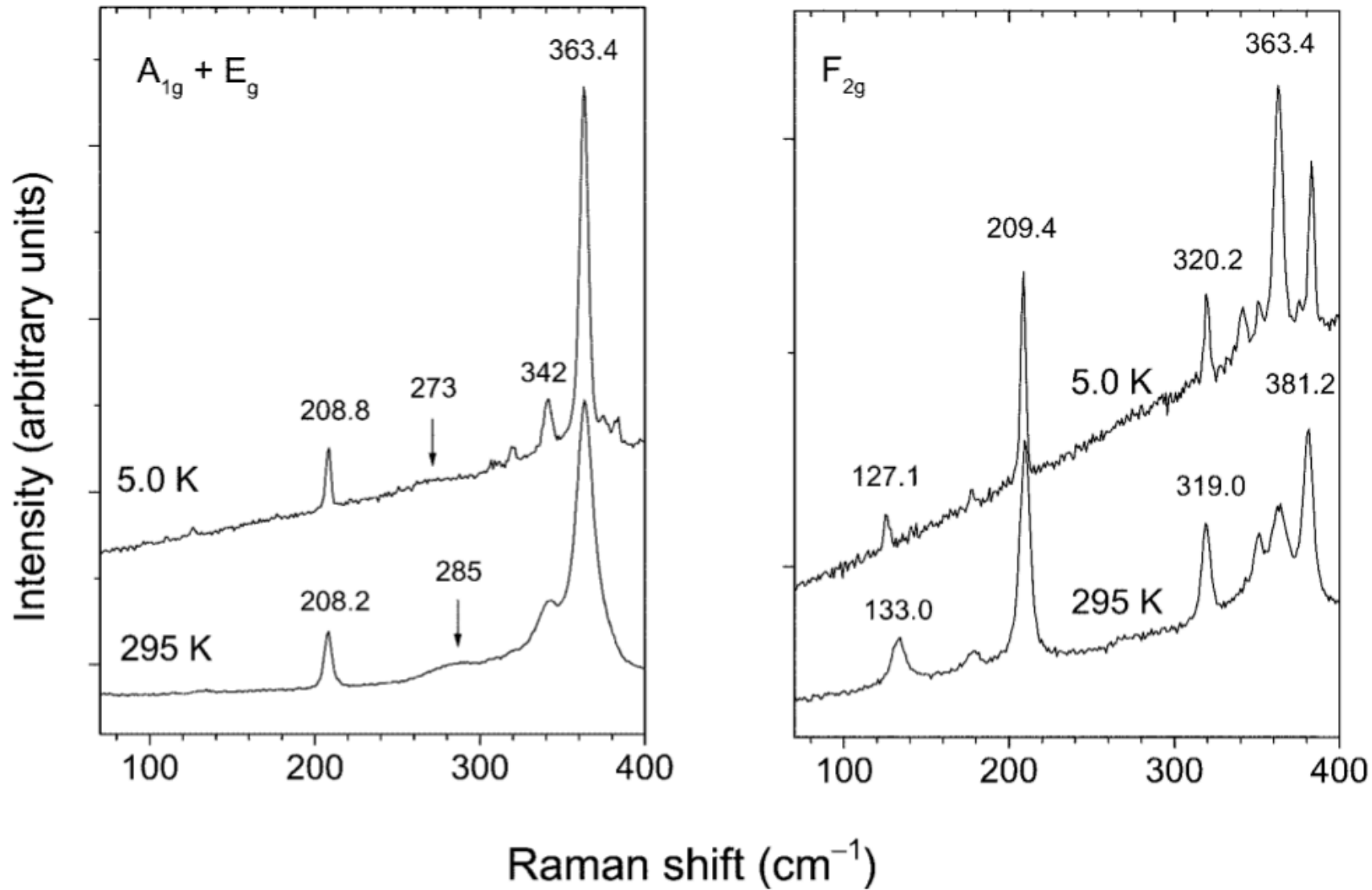
Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998		
Eg	-	309	308		
	342	-			
	365	379	364		
	439	439	431		
	911	867	816		
F2g	-	-	-		
	-	-	-		
	230	230			
	285	318	297		
	318	342	323		

Experiment

Simulation

Pyrope

Kolesov and Geiger 2000



Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	
Eg	-	309	308	285	
	342	-			
	365	379	364	-	
	439	439	431	-	
	911	867	816	-	
F2g	-	-	-	133	
	-	-	-	-	
	230	230		-	
	285	318	297	322	
	318	342	323	342	

Experiment

Simulation

Pyrope

To be continued...

2. Main Course - Theory



Response Properties

Electric
Field

Magnetic
Field

Nuclear
Spin

Atomic
Positions

$$\text{property} \propto \left(\frac{d}{dF} \right)^f \left(\frac{d}{dB} \right)^b \left(\frac{d}{dI} \right)^i \left(\frac{d}{dR} \right)^r E \Big|_0$$

f	b	i	r	property
1	0	0	0	electric dipole moment
0	1	0	0	magnetic dipole moment
0	0	1	0	hyperfine coupling constant
0	0	0	1	energy gradient
2	0	0	0	electric polarizability
0	2	0	0	magnetizability
0	0	2	0	spin–spin coupling (different nuclei)
0	0	0	2	harmonic vibrational frequencies
3	0	0	0	First Hyperpolarizability / SHG
...				...

Response Properties

Electric
Field

Magnetic
Field

Nuclear
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Atomic
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$$\text{property} \propto \left(\frac{d}{dF} \right)^f \left(\frac{d}{dB} \right)^b \left(\frac{d}{dI} \right)^i \left(\frac{d}{dR} \right)^r E \Big|_0$$

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...

...

Response Properties

Electric
Field

Magnetic
Field

Nuclear
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Positions

$$\text{property} \propto \left(\frac{d}{dF} \right)^f \left(\frac{d}{dB} \right)^b \left(\frac{d}{dI} \right)^i \left(\frac{d}{dR} \right)^r E \Big|_0$$

f	b	i	r	property
1	1	0	0	circular dichroism
1	0	0	1	infra-red absorption intensities
0	1	1	0	nuclear magnetic shielding
2	1	0	0	magnetic circular dichroism (Faraday effect)
2	0	0	1	Raman intensities
1	0	0	2	infra-red intensities for overtone and hot bands
2	2	0	0	Cotton–Mouton effect
2	0	0	2	Raman intensities for overtone and hot bands
...				...
...				...

Response Properties

Electric
Field

Magnetic
Field

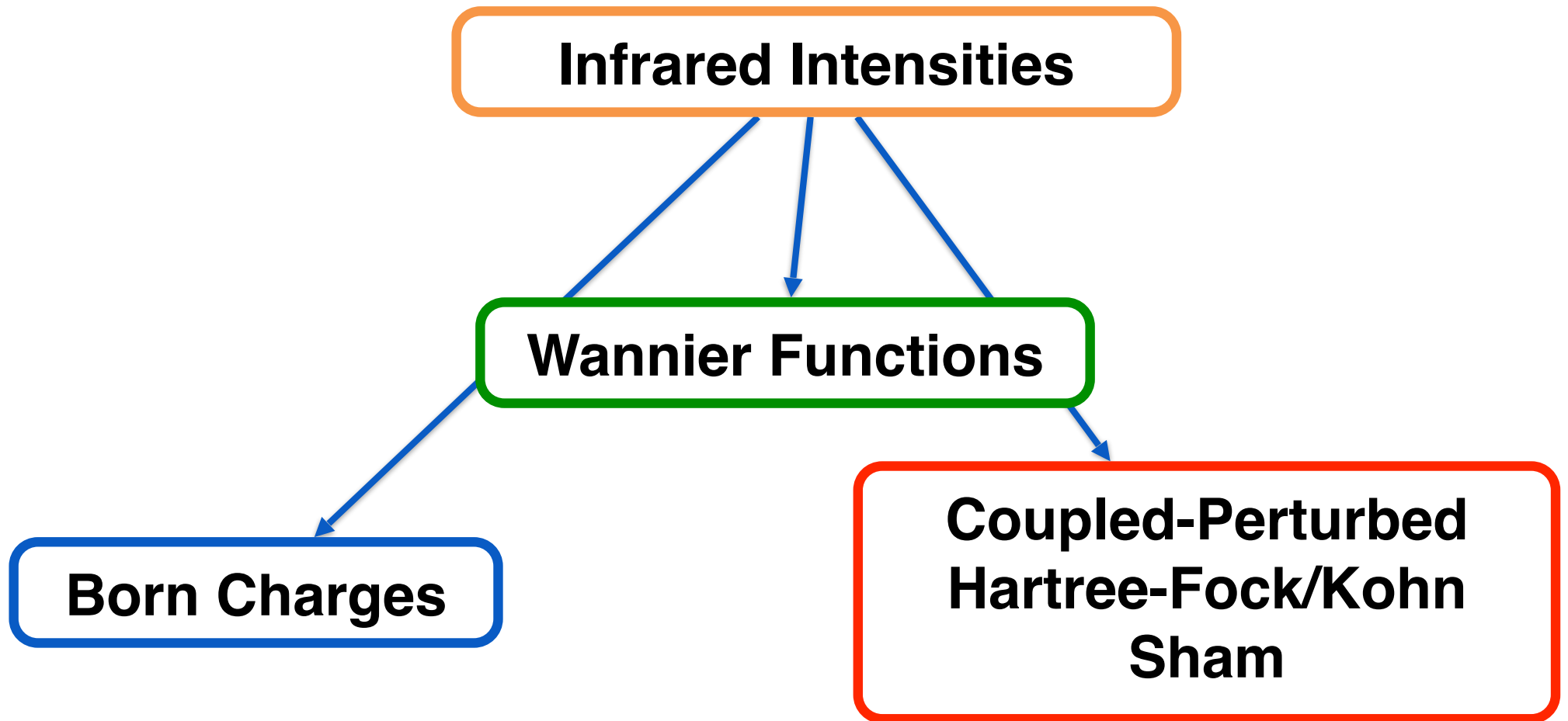
Nuclear
Spin

Atomic
Positions

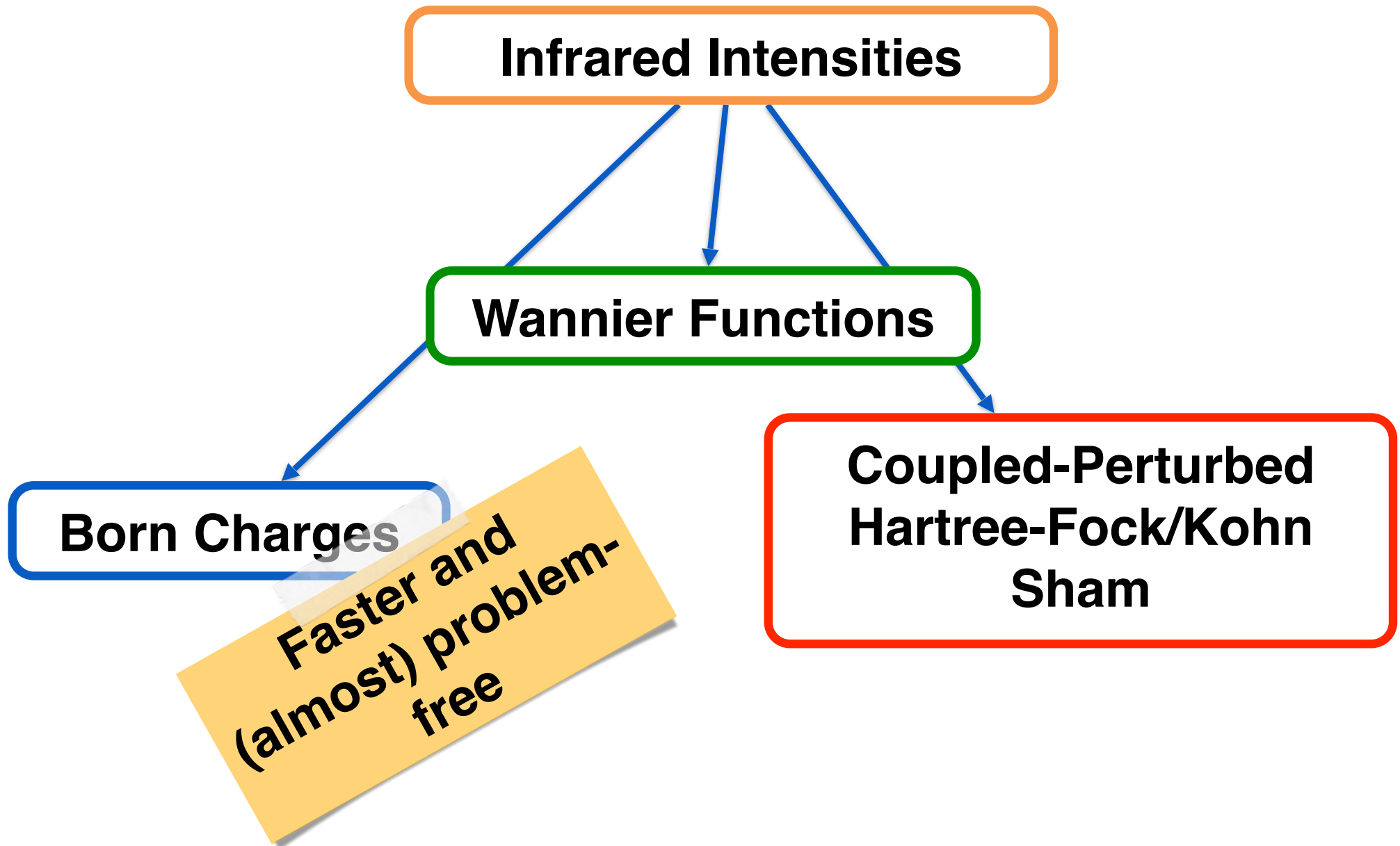
$$\text{property} \propto \left(\frac{d}{dF} \right)^f \left(\frac{d}{dB} \right)^b \left(\frac{d}{dI} \right)^i \left(\frac{d}{dR} \right)^r E \Big|_0$$

f	b	i	r	property
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2	0	0	2	Raman intensities for overtone and hot bands
...				...
...				...

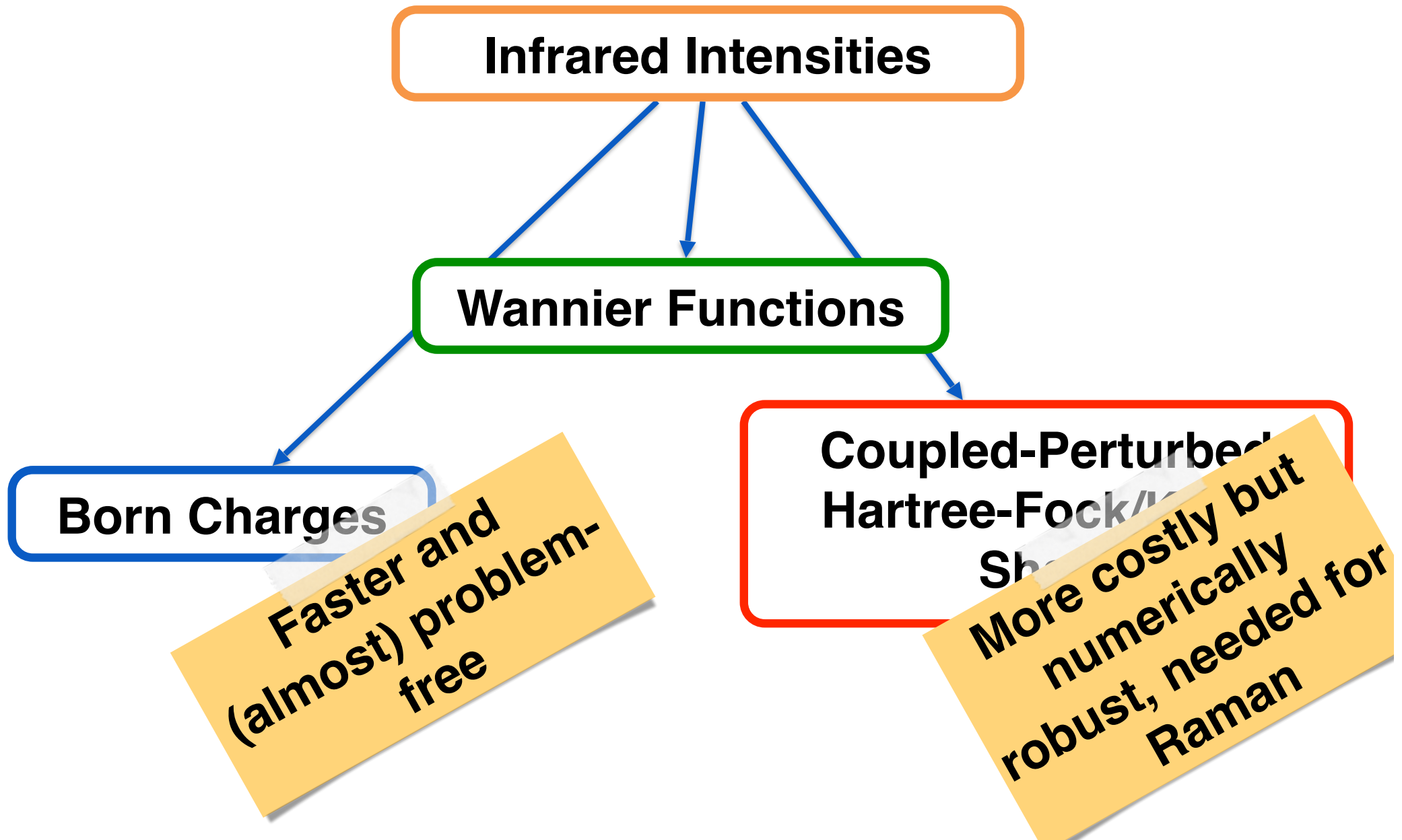
IR-Three possibilities



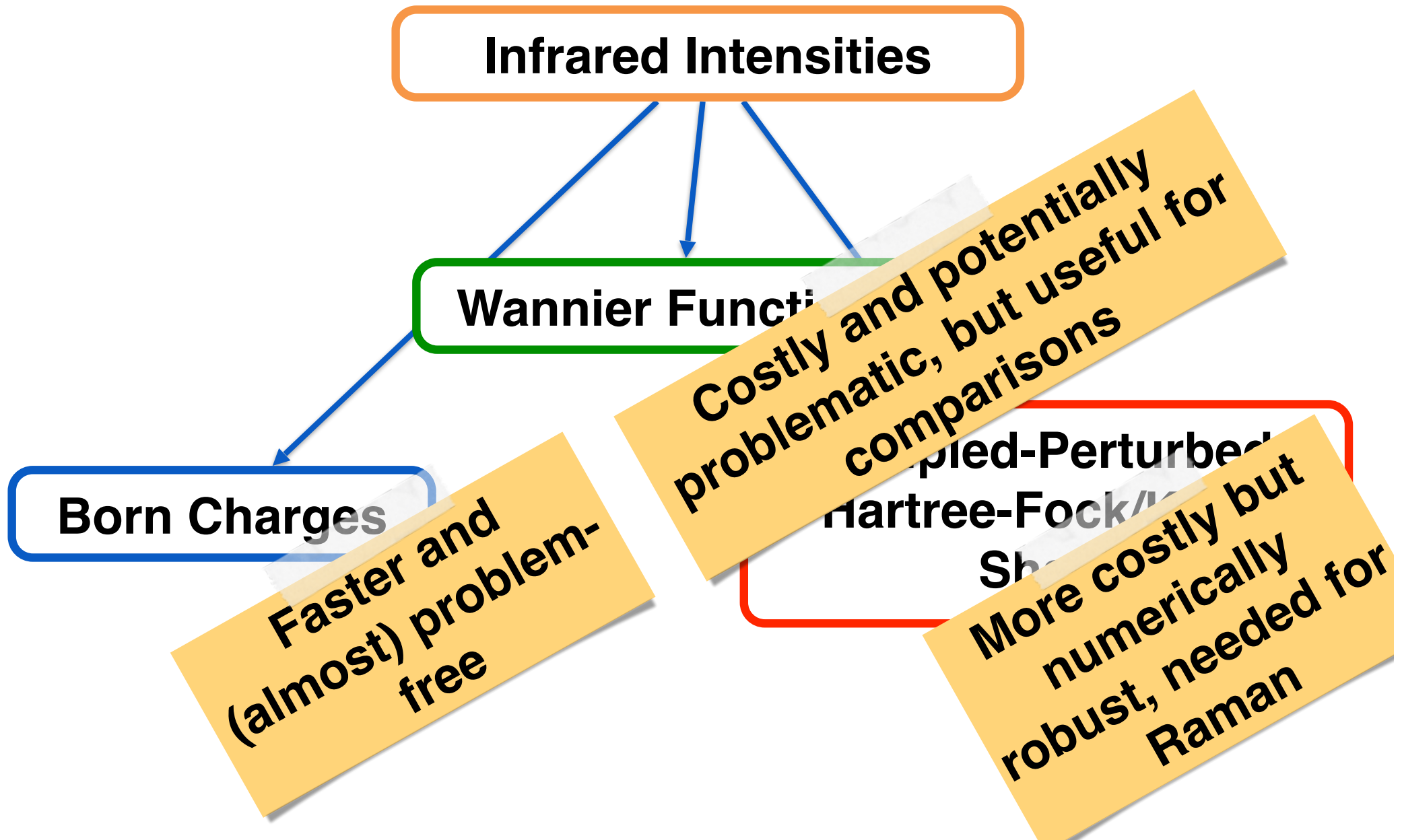
IR-Three possibilities



IR-Three possibilities



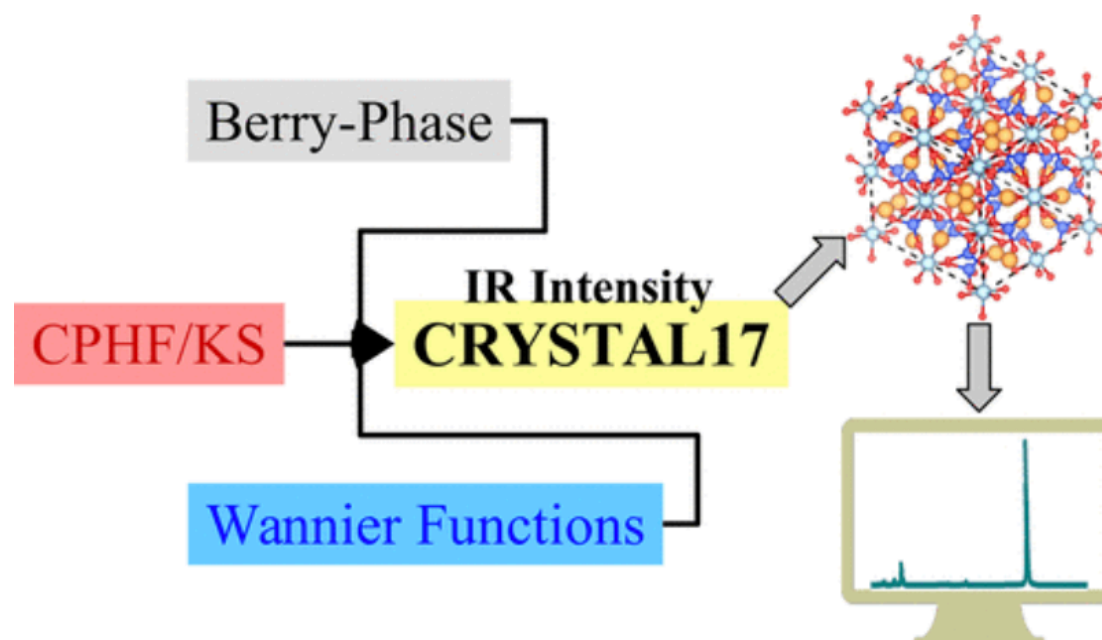
IR-Three possibilities



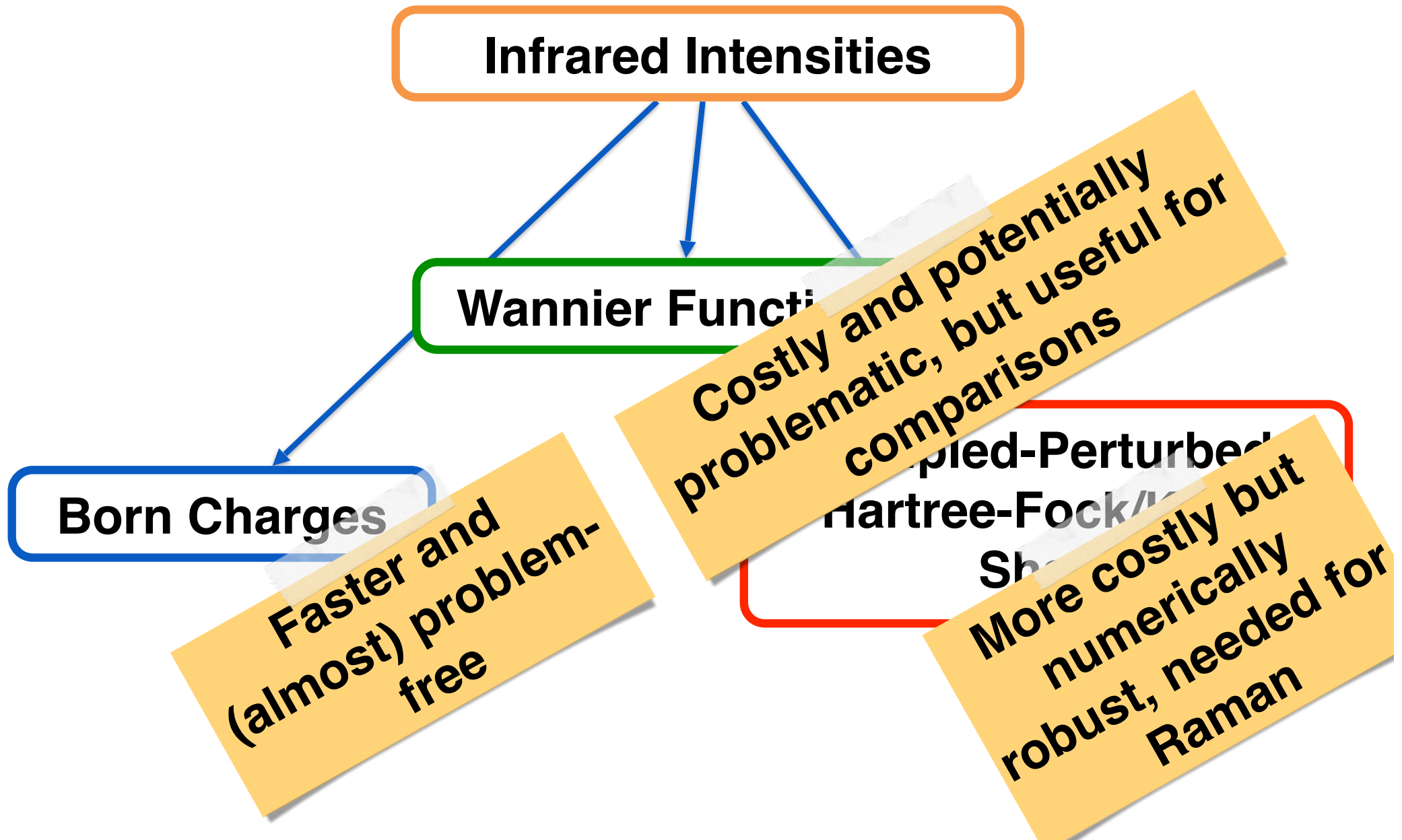
Different paths to the same goal

Calculation of the Infrared Intensity of Crystalline Systems. A Comparison of Three Strategies Based on Berry Phase, Wannier Function, and Coupled-Perturbed Kohn–Sham Methods

R. Dovesi,^{*,†,‡} B. Kirtman,[‡] L. Maschio,^{*,†,‡} J. Maul,^{†,‡} F. Pascale,[¶] and M. Rérat[§]

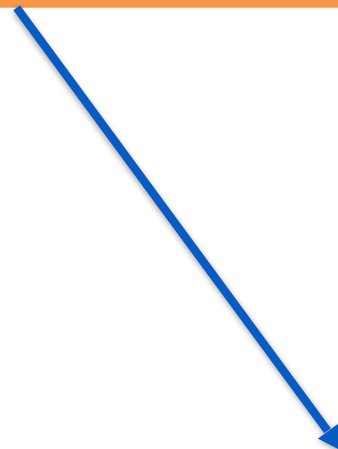


IR-Three possibilities



Raman - One possibility

Raman Intensities



**Coupled-Perturbed
Hartree-Fock/Kohn
Sham**

**Also generates
IR intensities as
a
by-product**

IR intensities

Born Charges are a key ingredient: derivative of the dipole moment w.r.t. atomic displacements:

how does the dipole moment changes when atoms move?

$$Z_{\alpha,ij}^* = \frac{\partial}{\partial u_{\alpha j}} \left(\frac{\partial E}{\partial \mathcal{E}_i} \right) \equiv \frac{\partial}{\partial u_{\alpha j}} \mu_i$$

$\vec{\mathcal{E}}$ = electric field

$\partial u_{\alpha j}$ = Atomic displacement

IR intensities

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they are composed by a nuclear and an electronic part

$$Z_{\alpha,ij}^*/V = \Delta P_{el}^{\alpha,ij} + \Delta P_{nuc}^{\alpha,ij}$$

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IR intensities

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they are composed by a nuclear and an electronic part

$$Z_{\alpha,ij}^*/V = \Delta P_{el}^{\alpha,ij} + \Delta P_{nuc}^{\alpha,ij}$$

the nuclear component is easily evaluated as:

$$Z_{\alpha,ij}^{*(nuc)} = \zeta_{\alpha} \delta_{ij}$$

ζ_{α} : atomic number of atom α .

IR intensities: Berry Phase

The Berry Phase approach involves a numerical derivative in reciprocal space:

$$\Delta \mathbf{P}_{el} = -\frac{i}{(2\pi)^3} \int d\mathbf{k} [\langle \Phi^{(1)}(\mathbf{k}) | \nabla_{\mathbf{k}} \Phi^{(1)}(\mathbf{k}) \rangle - \langle \Phi^{(0)}(\mathbf{k}) | \nabla_{\mathbf{k}} \Phi^{(0)}(\mathbf{k}) \rangle]$$

where

$$|\Phi^{(\lambda)}(\mathbf{k})\rangle = \frac{1}{\sqrt{n!}} |u_1^{(\lambda)}(\mathbf{k}) \bar{u}_1^{(\lambda)}(\mathbf{k}) \dots u_{n/2}^{(\lambda)}(\mathbf{k}) \bar{u}_{n/2}^{(\lambda)}(\mathbf{k})|$$

and $u_i(\bar{u}_i)$ is the periodic part of the i th HF occupied spin $\alpha(\beta)$ crystalline orbital in the Bloch form:

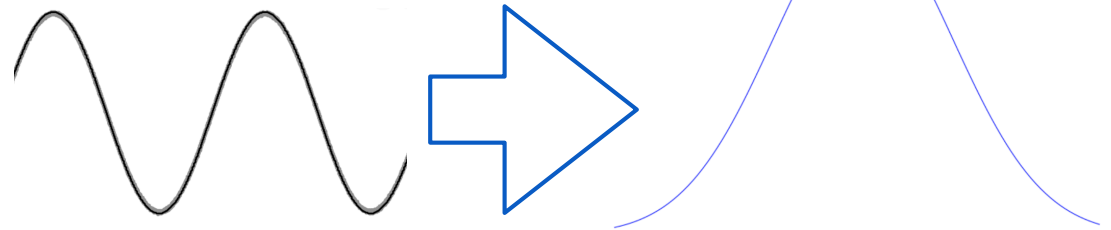
$$\psi_i^{(\lambda)}(\mathbf{r}, \mathbf{k}) = e^{i\mathbf{k}\cdot\mathbf{r}} u_i^{(\lambda)}(\mathbf{r}, \mathbf{k})$$

IR intensities: Wannier Approach

The Wannier Function approach involves a numerical derivative in direct space space:

First step involves a localization of the SCF solution

$$\omega_i(\mathbf{r}) = \sum_{\mu=1}^M \sum_{\mathbf{g}} c_{\mu i}^{\mathbf{g}} \varphi_{\mu}(\mathbf{r} - \mathbf{s}_{\mu} - \mathbf{R}_{\mathbf{g}})$$



Then centroids of the Wannier Functions are evaluated

$$\bar{\mathbf{r}}_i = \int d\mathbf{r} \, \mathbf{r} |\omega_i(\mathbf{r})|^2,$$

And the $\Delta \mathbf{P}_{\text{el}}$ is evaluated as a distance between centroids in the central and distorted positions

Analytical Derivatives: CPHF approach

Born Charges (IR intensities): derivative of the dipole moment

$$Z_{aA,b}^* = \left. \frac{\partial^2 E^{TOT}}{\partial \mathcal{R}_a^A \partial \mathcal{E}_b} \right|_{\mathcal{E}=0, \mathcal{R}_0} = \frac{\partial \mu_b}{\partial \mathcal{R}_a^A}$$

Within Placzek approximation, **Raman tensor elements** are defined as the derivative of polarizability :

$$\left. \frac{\partial \alpha_{b,c}}{\partial \mathcal{R}_a^A} \right|_{\mathcal{R}_0} = \left. \frac{\partial^3 E^{TOT}}{\partial \mathcal{R}_a^A \partial \mathcal{E}_b \partial \mathcal{E}_c} \right|_{\mathcal{E}=0, \mathcal{R}_0}$$

$\vec{\mathcal{E}}$ = electric field

$\mathcal{R}_a^A$ = Atomic displacement

Electric Field in Periodic Systems

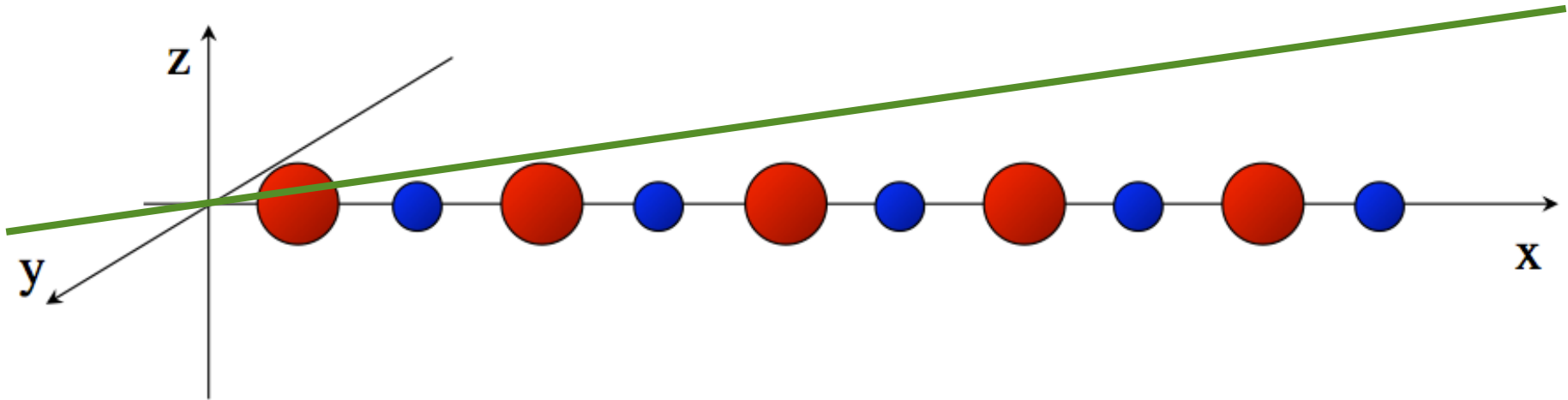
$$\hat{V}(\vec{r}) = \vec{\mathcal{E}} \cdot \vec{r}$$

This operator is not consistent with the periodic boundary conditions, it is not bound and breaks the translational invariance of the system.

Electric Field in Periodic Systems

$$\hat{V}(\vec{r}) = \vec{\mathcal{E}} \cdot \vec{r}$$

This operator is not consistent with the periodic boundary conditions, it is not bound and breaks the translational invariance of the system.



The total energy of a crystal in an electric field

The total energy (E_{tot}) of a crystal (or a molecule) in a “weak” electric field (F) can be expressed as a perturbative series of the field components plus the total energy of the field-free system (E_{tot}^0):

$$E_{tot}(\mathbf{F}) = E_{tot}^0 - \boldsymbol{\mu} \cdot \mathbf{F} - \frac{1}{2!} \alpha F^2 - \frac{1}{3!} \beta F^3 - \frac{1}{4!} \gamma F^4 - \dots$$

$$\mu_t = - \left(\frac{\partial E_{tot}}{\partial F_t} \right)_{\varepsilon=0} \quad \text{dipole moment}$$

$$\alpha_{tu} = - \left(\frac{\partial^2 E_{tot}}{\partial F_t \partial F_u} \right)_{\varepsilon=0} \quad \text{polarizability}$$

$$\beta_{tuv} = - \left(\frac{\partial^3 E_{tot}}{\partial F_t \partial F_u \partial F_v} \right)_{\varepsilon=0} \quad \text{first-order hyperpolarizability}$$

$$\gamma_{tuvw} = - \left(\frac{\partial^4 E_{tot}}{\partial F_t \partial F_u \partial F_v \partial F_w} \right)_{\varepsilon=0} \quad \text{second-order hyperpolarizability}$$

The effect of a low-intensity high-frequency electric field (F) applied to a crystal within the periodic boundary conditions can be represented by the following perturbative term in the Hamiltonian operator:

$$\Omega(\mathbf{F}, \mathbf{k}) = \mathbf{F} \cdot [\mathbf{r} + i \nabla_{\mathbf{k}}]$$

position operator

gradient in $\mathbf{k}$ space

Ω depends on $\mathbf{k}$,
any point in the
reciprocal space

Steps of the IR-Raman intensity Algorithm

$$Z_{aA,b}^* = \frac{\partial^2 E^{TOT}}{\partial \mathcal{R}_a^A \partial \mathcal{E}_b} \Big|_{\mathcal{E}=0, \mathcal{R}_0} = \frac{\partial \mu_b}{\partial \mathcal{R}_a^A} \quad \frac{\partial \alpha_{b,c}}{\partial \mathcal{R}_a^A} \Big|_{\mathcal{R}_0} = \frac{\partial^3 E^{TOT}}{\partial \mathcal{R}_a^A \partial \mathcal{E}_b \partial \mathcal{E}_c} \Big|_{\mathcal{E}=0, \mathcal{R}_0}$$

- 1) One CPHF calculation
- 2) One CPHF2 calculation (only for Raman)
- 3) Integral gradients at the equilibrium geometry.

IR and Raman tensors are built assembling all these ingredients and then contracted with eigenmodes.

Second Harmonic Generation and Pockels

We get the frequency-dependent hyperpolarizability tensor from 1st-order CPHF, but we need **3 frequencies**, hence to converge (at most) three CPHF procedures

$$\beta_{tuv}(-\omega_\sigma; \omega_1, \omega_2) = -\frac{2}{n_k} \mathcal{K}_{-\omega_\sigma, \omega_1, \omega_2} \sum_{\vec{k}} \Re \sum_i \sum_a P_{\frac{t}{-\omega_\sigma}, \frac{u}{+\omega_1}, \frac{v}{+\omega_2}} \\ \times \left\{ U_{ai(-\omega_\sigma)}^{(t)*}(\vec{k}) \left[\sum_b G_{ab(+\omega_1)}^{(u)}(\vec{k}) U_{bi(+\omega_2)}^{(v)}(\vec{k}) - \sum_j U_{aj(+\omega_2)}^{(v)}(\vec{k}) G_{ji(+\omega_1)}^{(u)}(\vec{k}) + i \frac{\partial U_{ai(+\omega_2)}^{(v)}(\vec{k})}{\partial k_u} \right] \right\}$$

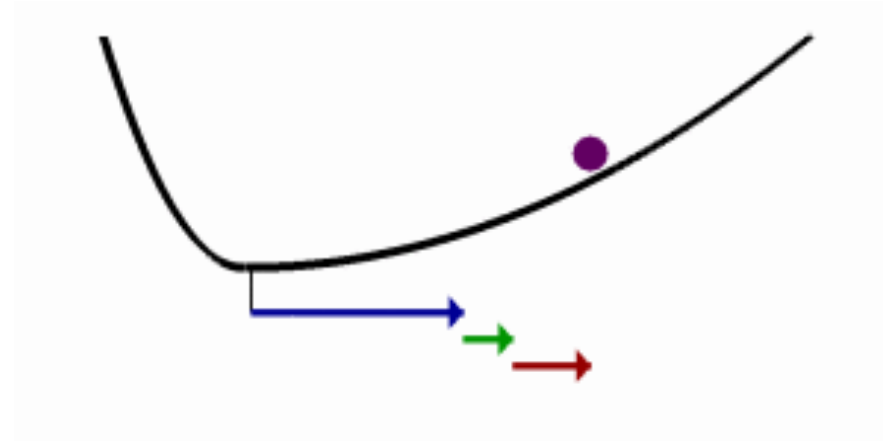
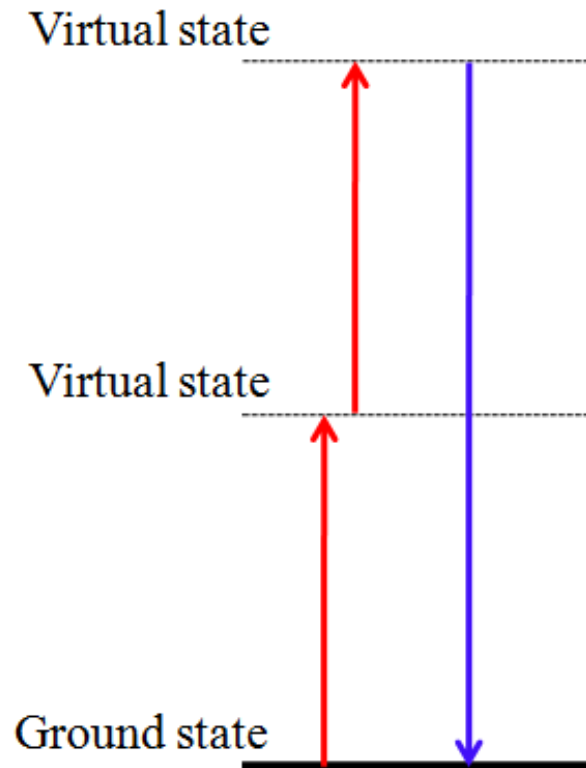
Dynamic first hyperpolarizability

$\beta(-\omega_\sigma; \omega_1, \omega_2)$ sum of frequencies must be zero.

Dynamic first hyperpolarizability

$$\beta(-\omega_\sigma; \omega_1, \omega_2)$$

$$\beta(-2\omega; \omega, \omega) \quad \text{Second-Harmonic Generation (SHG)}$$



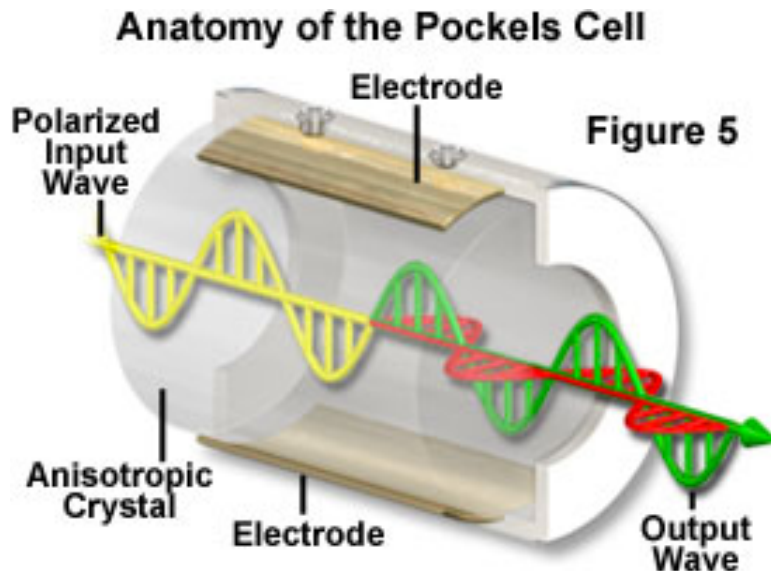
Blue arrow: ordinary (linear) susceptibility
Green arrow: second-harmonic generation
Red arrow: optical rectification.

Dynamic first hyperpolarizability

$$\beta(-\omega_\sigma; \omega_1, \omega_2)$$

$$\beta(-2\omega; \omega, \omega) \quad \text{Second-Harmonic Generation (SHG)}$$

$$\beta(-\omega; \omega, 0) \quad \text{Pockels Effect}$$



Pockels Cells are used to rotate the polarization of a passing beam.

3. Cheese - from theory to experiment



IR Intensities

The IR intensity of the p-th mode: $A_p \propto d_p \overline{Z}_{p,j}^2$

Reflectivity is calculated from dielectric constant by means of:

$$R(\nu) = \left| \frac{\sqrt{\epsilon(\nu) - \sin^2 \theta} - \cos \theta}{\sqrt{\epsilon(\nu) - \sin^2 \theta} + \cos \theta} \right|^2$$

(θ is the beam incident angle)

The **dielectric function** is obtained with the classical dispersion relation:

$$\epsilon(\nu) = \epsilon_\infty + \sum_j \frac{f_j \nu_j^2}{\nu_j^2 - \nu^2 - i\nu\gamma_j} \qquad f_j = \frac{4\pi}{\Omega} \frac{\overline{Z}_j^2}{\nu_j^2}$$

Raman Intensities- Single Crystal

$$I_{xy} \sim C\alpha_{xy}^2$$

where the prefactor C is

$$C \sim (\omega_L - \omega_i)^4 \frac{1 + n(\omega_i)}{30\omega_i}$$

The temperature dependence is given by the Bose occupancy factor, $n(\omega)$:

$$1 + n(\omega_i) = \left[1 - \exp\left(\frac{-\hbar\omega_i}{kT}\right) \right]^{-1}$$

The Raman tensors are defined as:

$$\alpha_{xy} = \sqrt{\Omega} \sum_{\alpha\tau} \frac{\partial \chi_{xy}}{\partial r_{\alpha\tau}}$$

Raman Intensities- Powder

Tensor invariants are obtained averaging the Raman directional intensities

$$G^{(0)} = \frac{\sum_{ii, i=x,y,z} (\alpha_{ii})^2}{3}$$

$$G^{(1)} = \frac{\sum_{ij, i,j=x,y,z} (\alpha_{ij} - \alpha_{ji})^2}{2}$$

$$G^{(2)} = \sum_{ij, i,j=x,y,z} \frac{(\alpha_{ij} + \alpha_{ji})^2}{2} + \frac{(\alpha_{ii} - \alpha_{jj})^2}{3}$$

$$I_{\parallel}^{\text{powder}} = C(10G^{(0)} + 4G^{(2)})$$

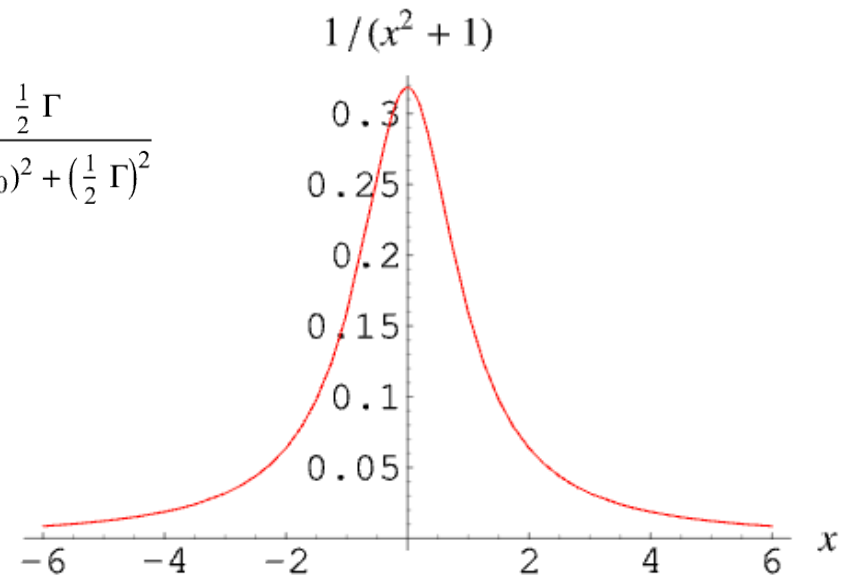
$$I_{\perp}^{\text{powder}} = C(5G^{(1)} + 3G^{(2)})$$

$$I_{\text{tot}}^{\text{powder}} = I_{\parallel}^{\text{powder}} + I_{\perp}^{\text{powder}}$$

Raman Intensities- Peak shape

A **Lorentzian Function** with fixed width and height proportional to the intensity is used to draw the Raman spectrum profile.

$$L(x) = \frac{1}{\pi} \frac{\frac{1}{2} \Gamma}{(x - x_0)^2 + \left(\frac{1}{2} \Gamma\right)^2}$$



What you should be aware of:

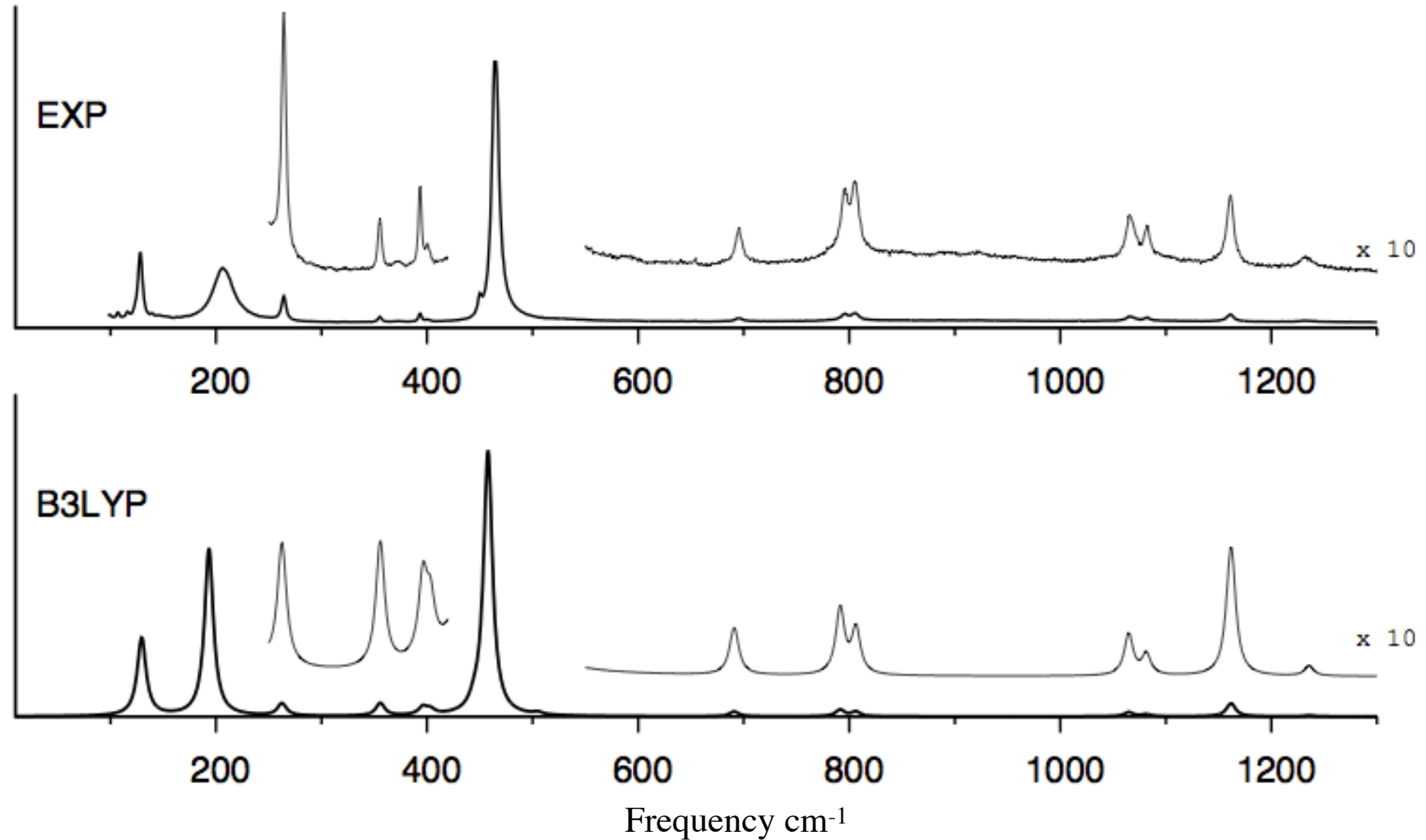
- 1) The width in reality **changes** for every peak (but it is very difficult and costly to compute).
- 2) The **intensity** as a physical quantity is actually the **area** of the peak, not its height.

4. Dessert - simulated spectra



Theory Vs Experiment: SiO₂

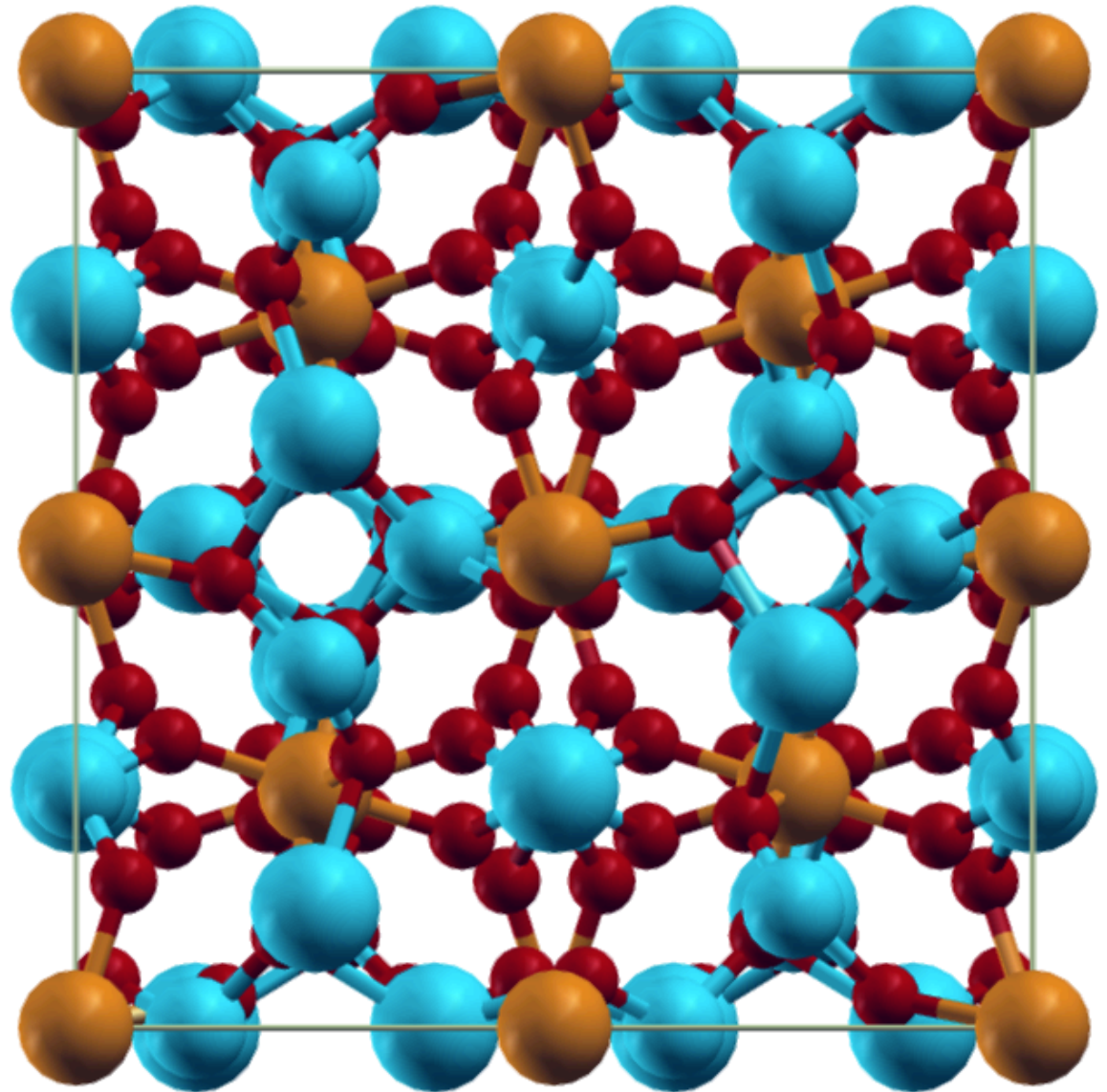
EXP: Handbook of Minerals Raman Spectra database of Lyon ENS



Pyrope

Garnets are important
rock-forming silicates

Pyrope: $\text{Mg}_3\text{Al}_2\text{Si}_3\text{O}_{12}$



Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
	439	439	431	-	
	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

data in cm⁻¹

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
	439	439	431	-	
	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

**Leakage
from F2g**

data in cm⁻¹

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
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	365	379	364	-	377
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F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

Confirmed

data in cm⁻¹

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
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	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

Confirmed

data in cm⁻¹

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
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	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

Nothing even
close!

377-assigned
527-assigned

data in cm⁻¹

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
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F2g	-	-	-	133	135
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	230	230		-	
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	318	342	323	342	

Confirmed

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
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F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

Confirmed

Experiment

Simulation

Pyrope

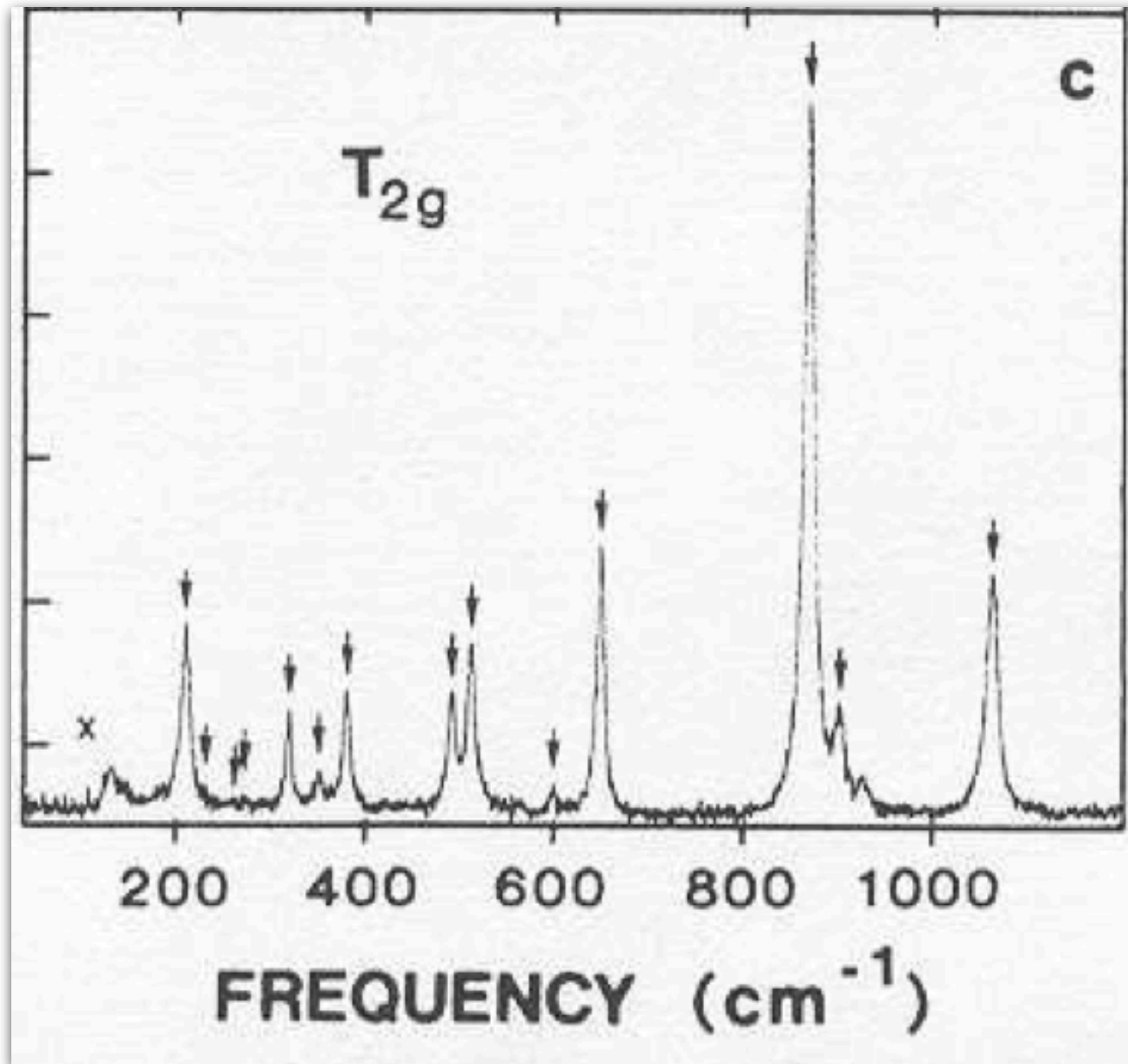
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Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
	439	439	431	-	
	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

**Never
reported**

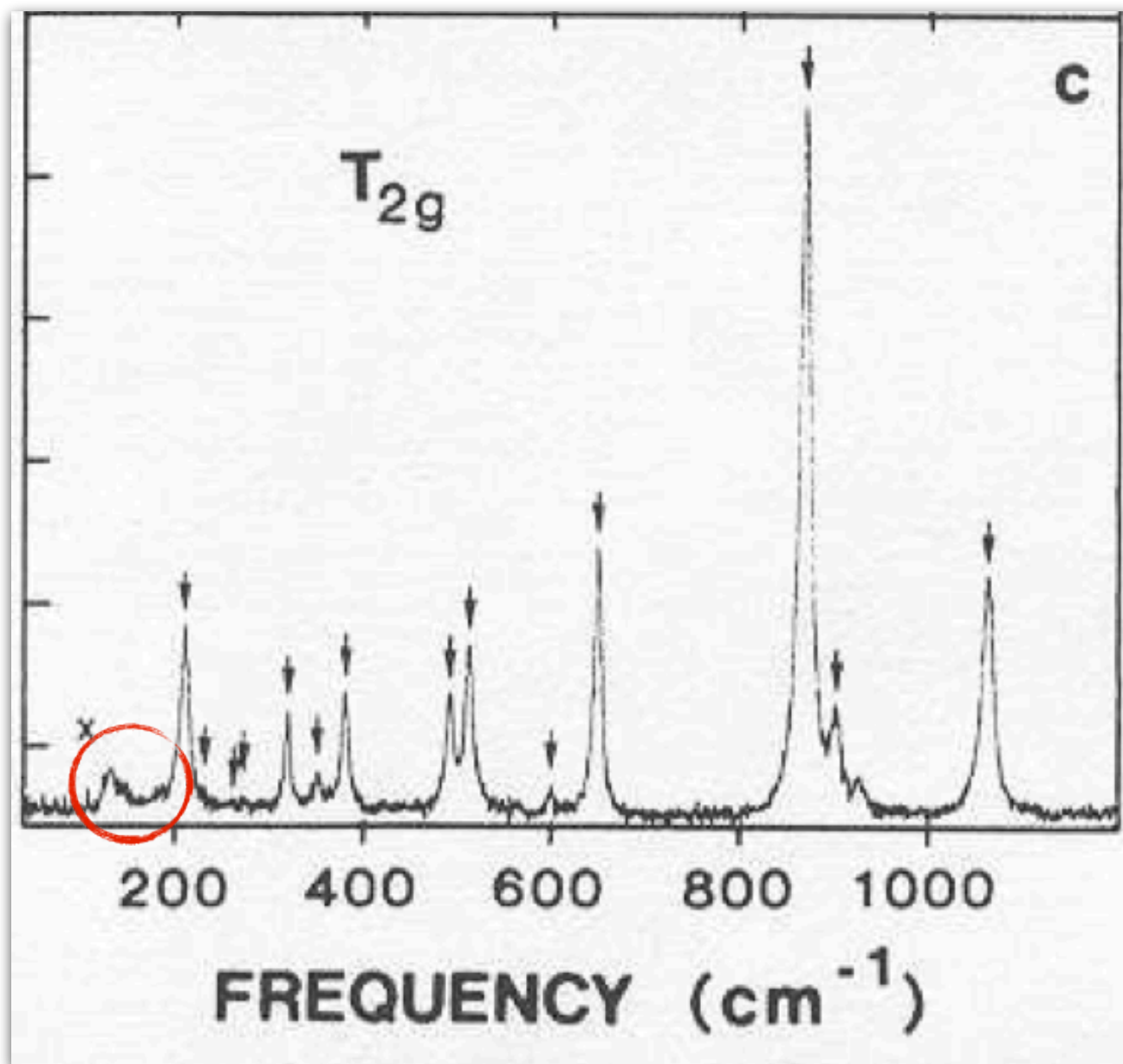
Experiment

Simulation

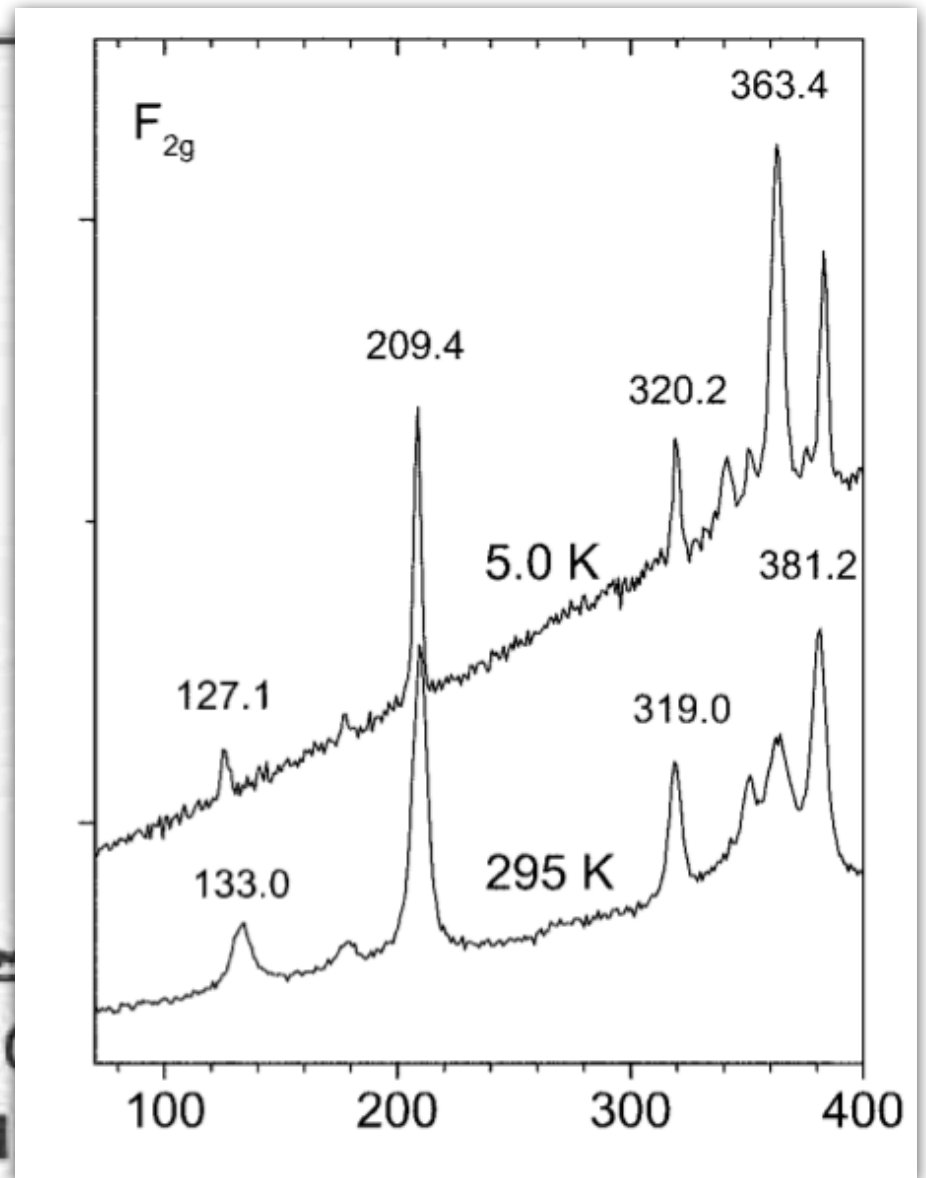
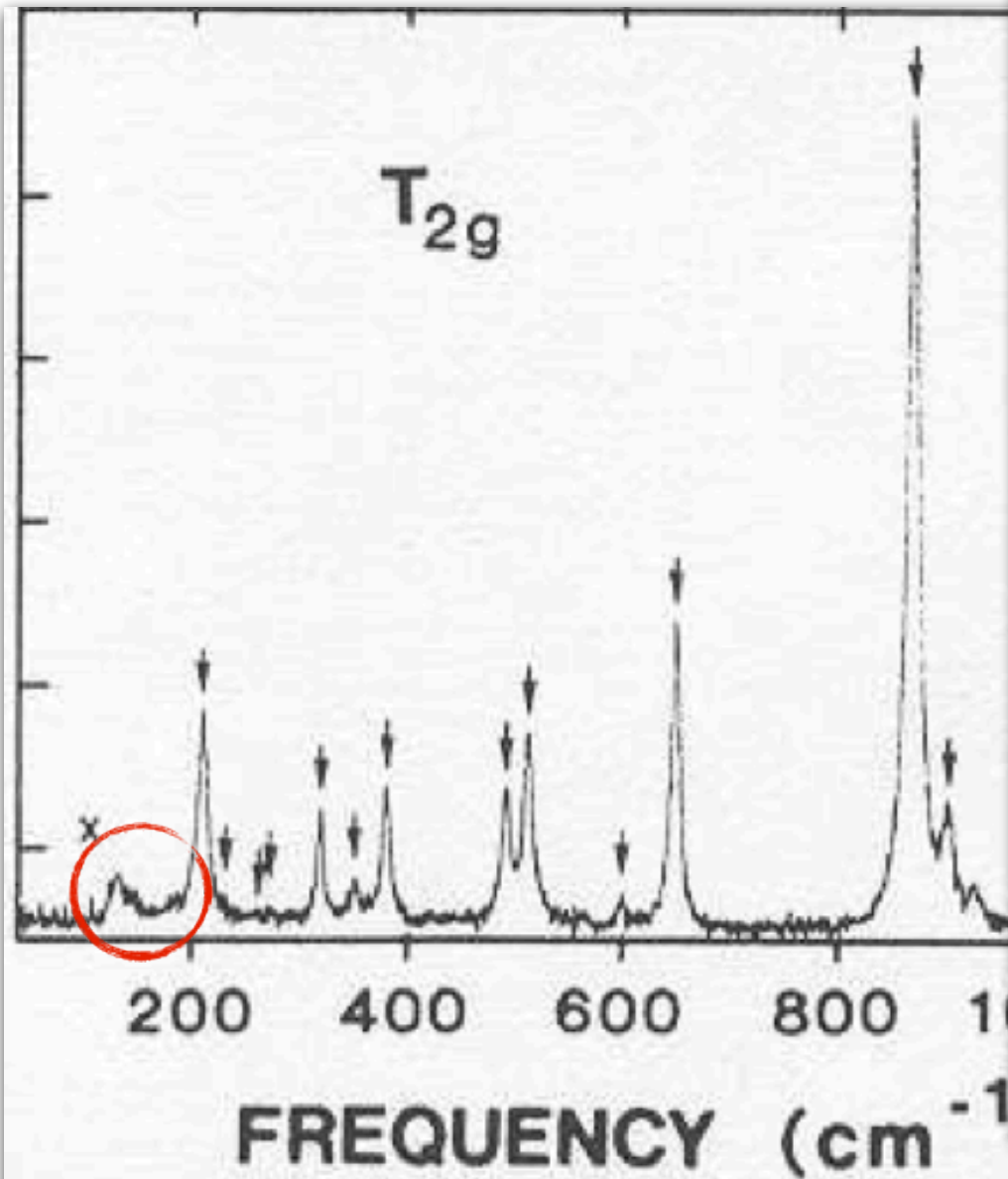
Pyrope



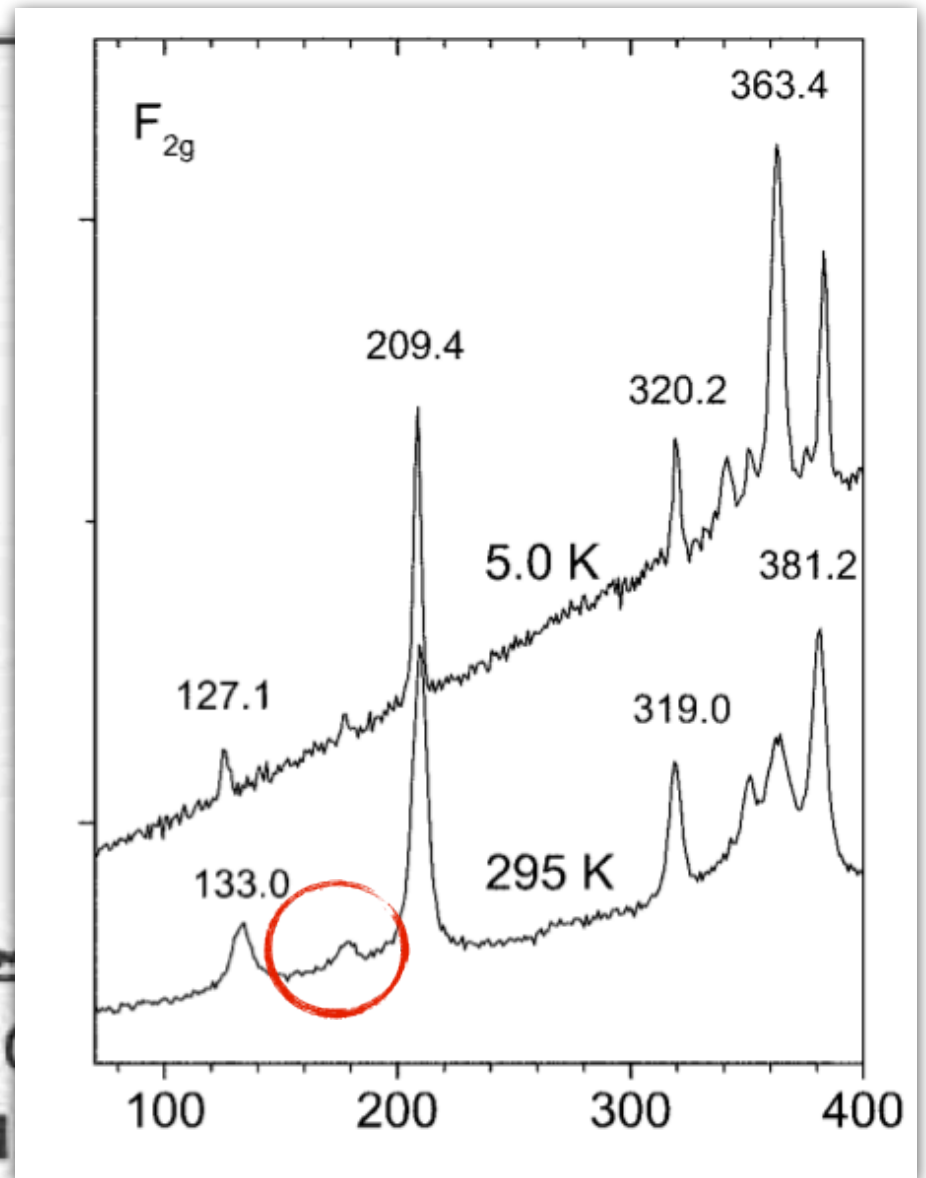
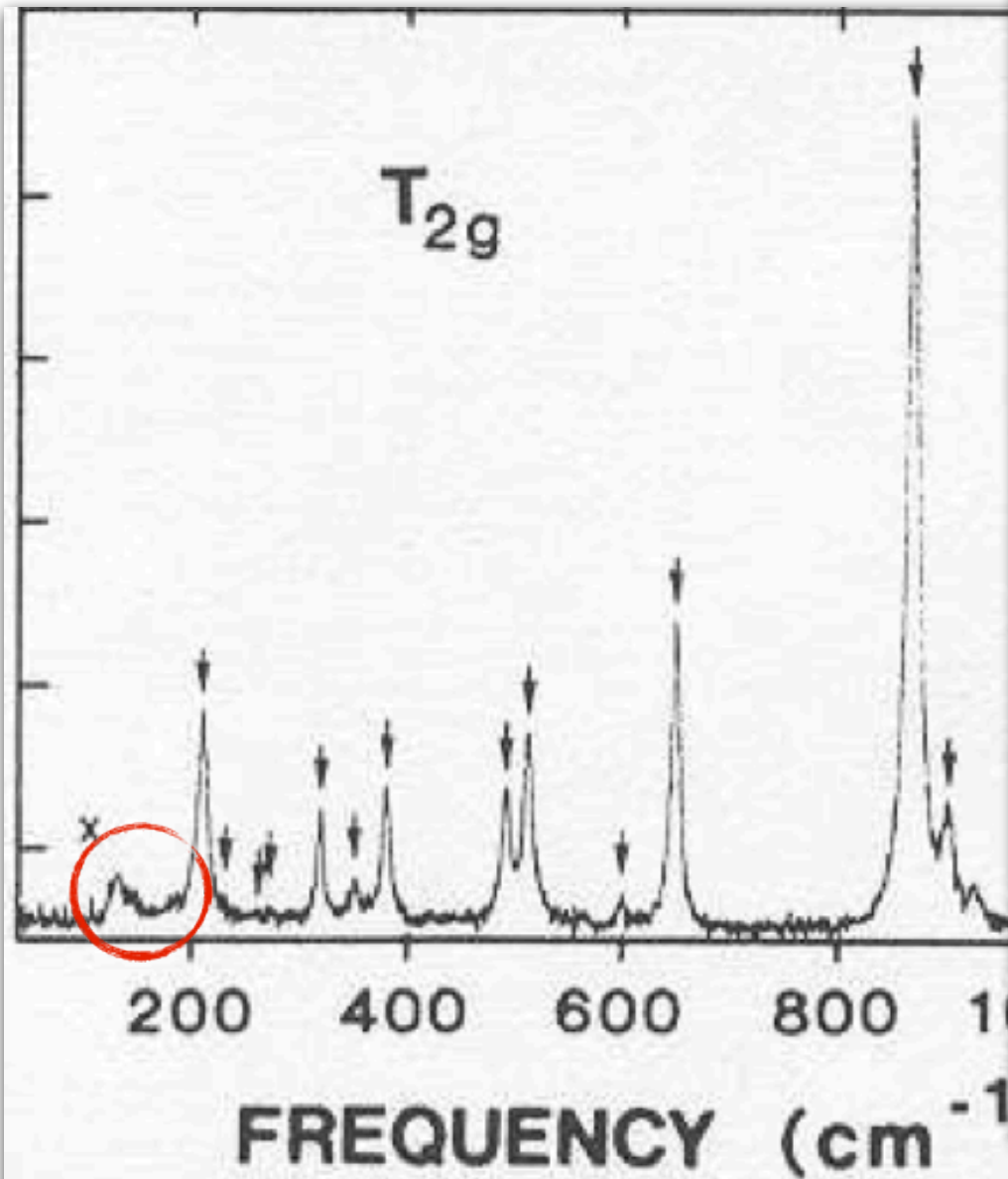
Pyrope



Pyrope



Pyrope



Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
	439	439	431	-	
	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

Never seen

Experiment

Simulation

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
	439	439	431	-	
	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	



Not confirmed

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
	439	439	431	-	
	911	867	816	-	864
F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
	285	318	297	322	319
	318	342	323	342	

“Confirmed”

Pyrope

Sym.	Hofmeister - Chopelas 1991	Chopelas et al. 1998	Chaplin et al. 1998	Kolesov - Geiger 2000	This Work 2013
Eg	-	309	308	285	308
	342	-			336
	365	379	364	-	377
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F2g	-	-	-	133	135
	-	-	-	-	181
	230	230		-	
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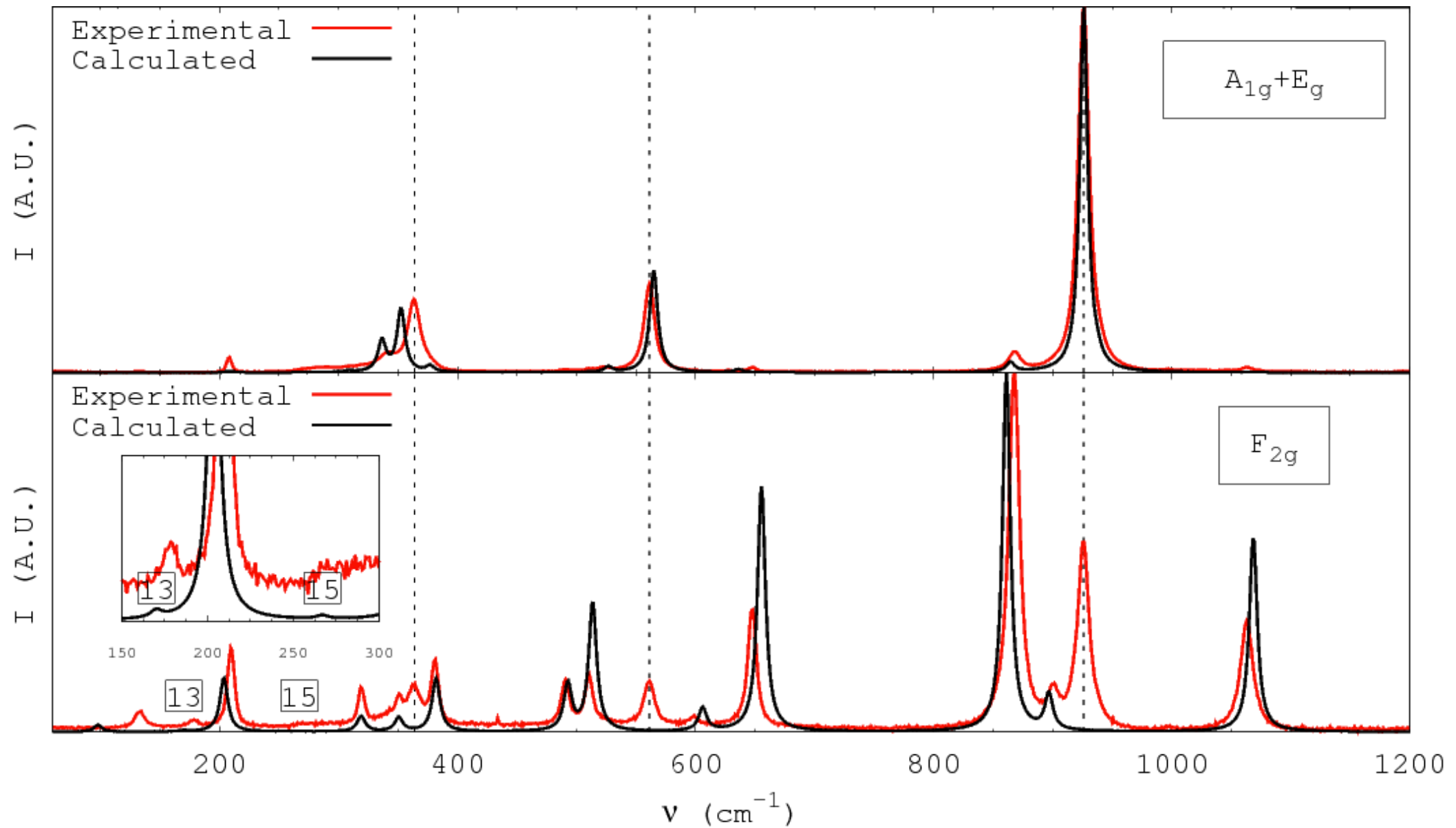
Leakage?

Pyrope

Some modes, though Raman active by symmetry considerations, have nearly zero intensity.

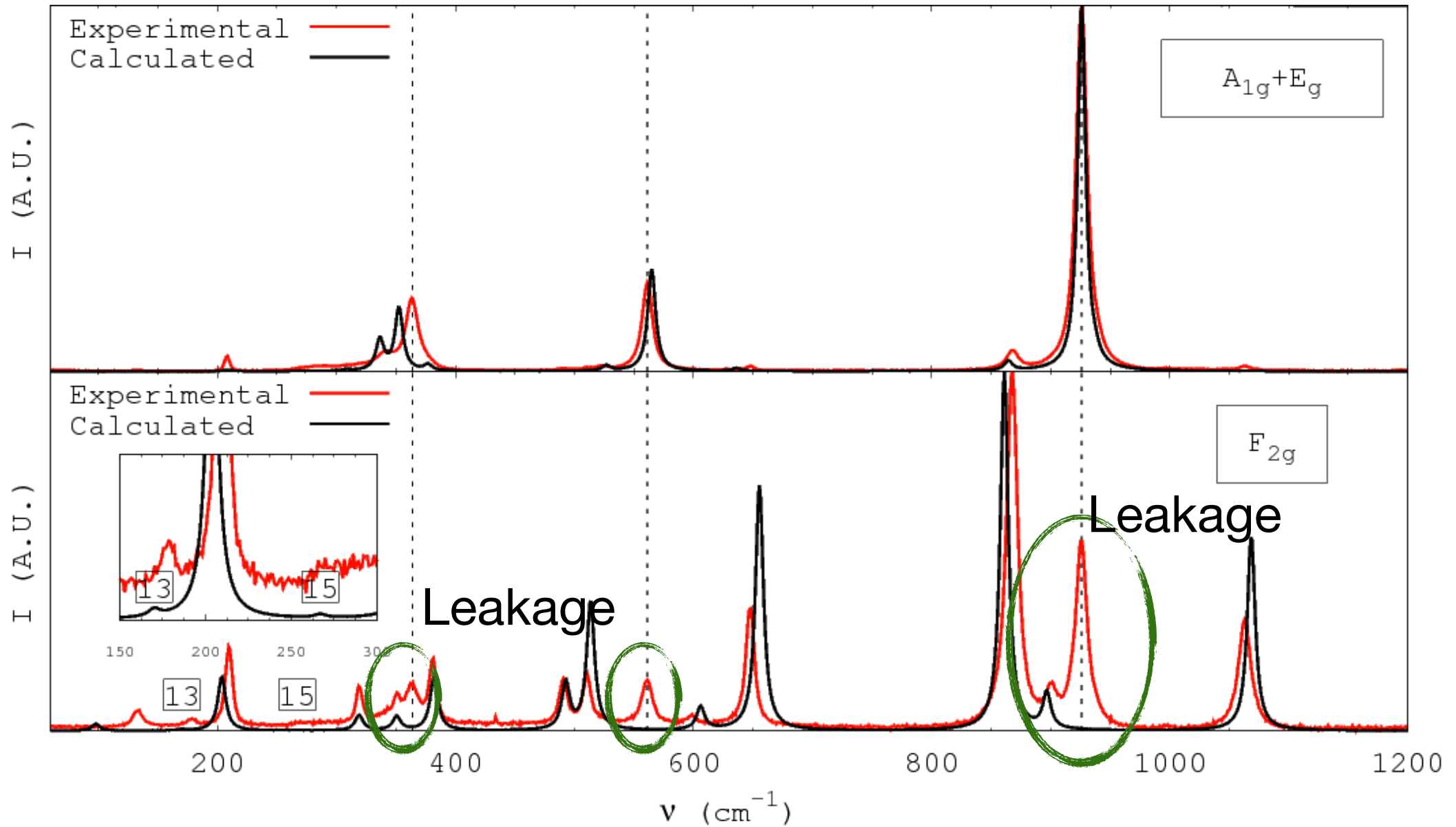
Assignment of experimental peaks is widely guided by experience

Pyrope



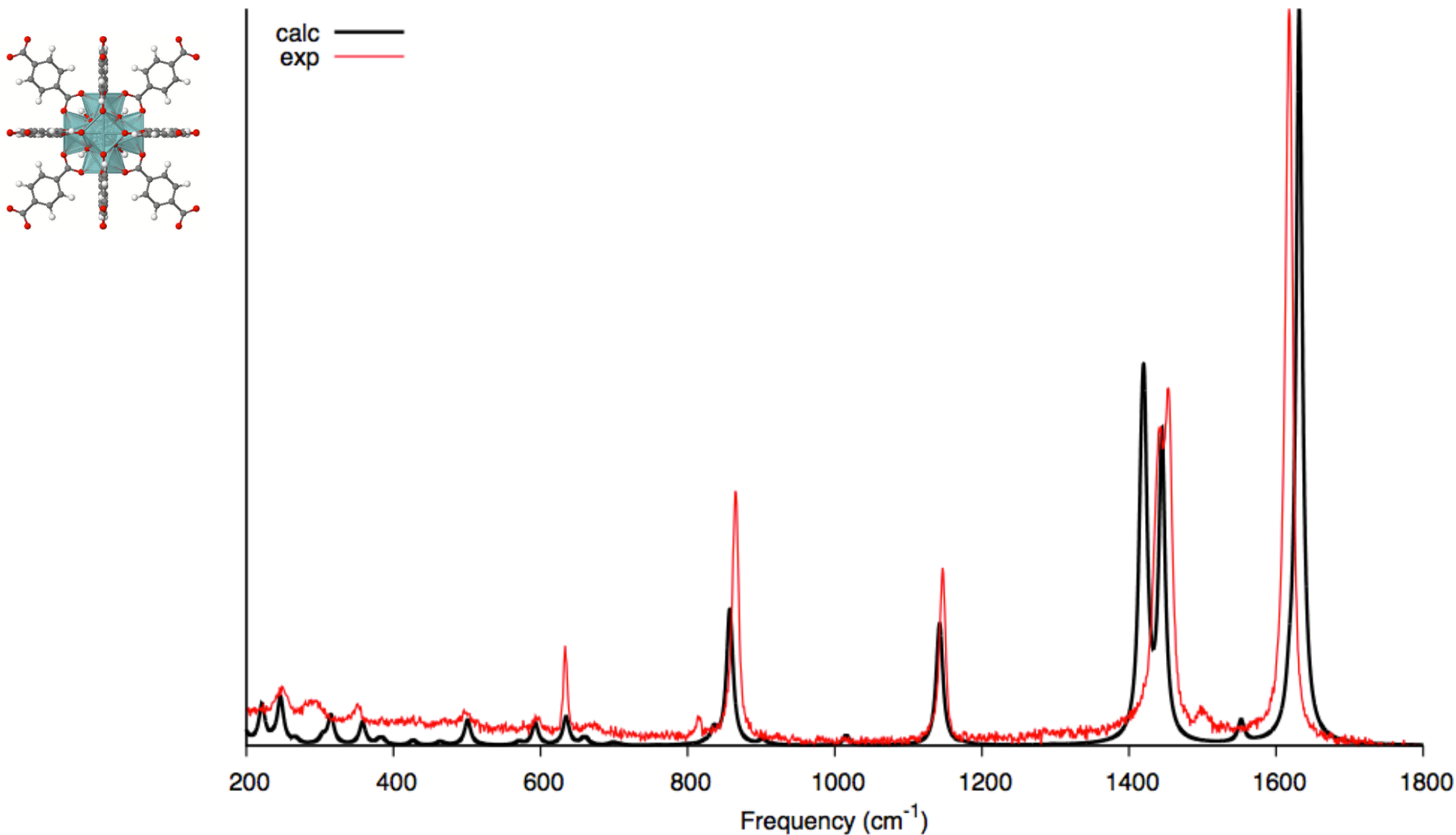
Experimental=Kolesov (2000)

Pyrope



UiO-66 Metal Organic Framework

More than 90 Raman-active modes

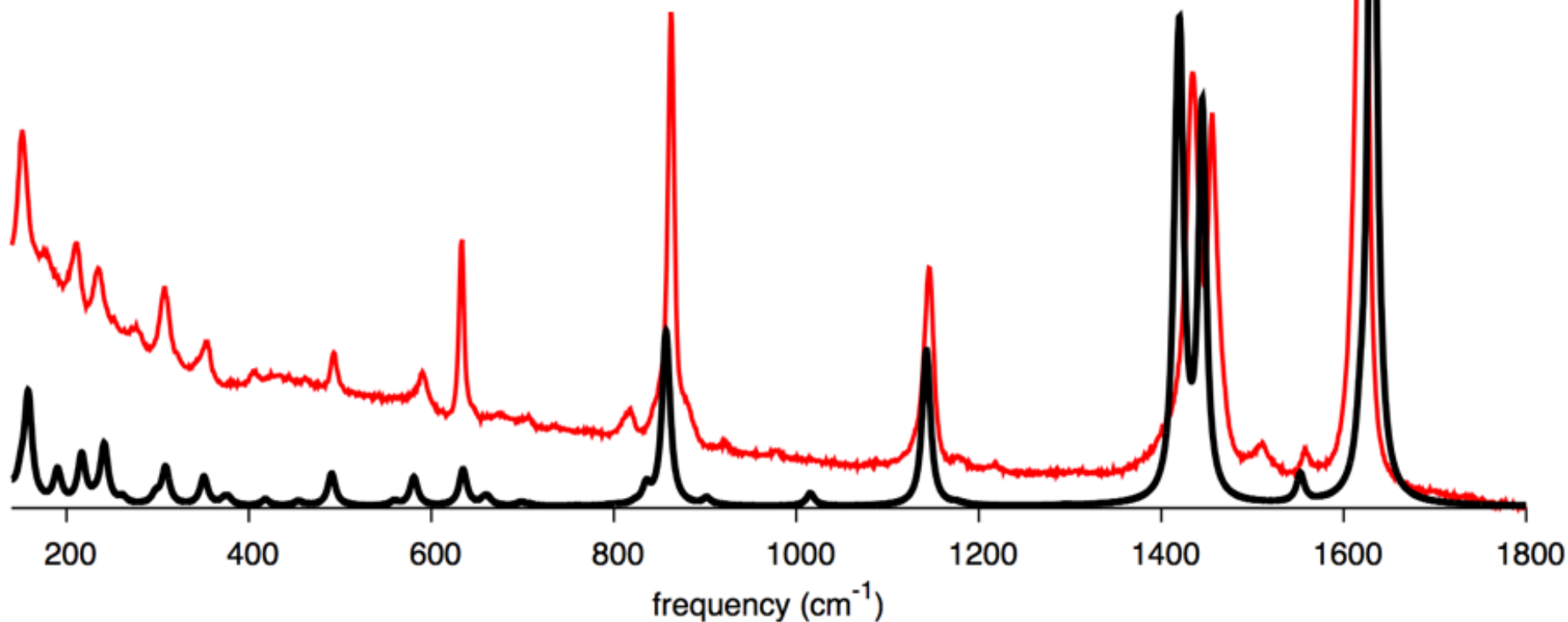
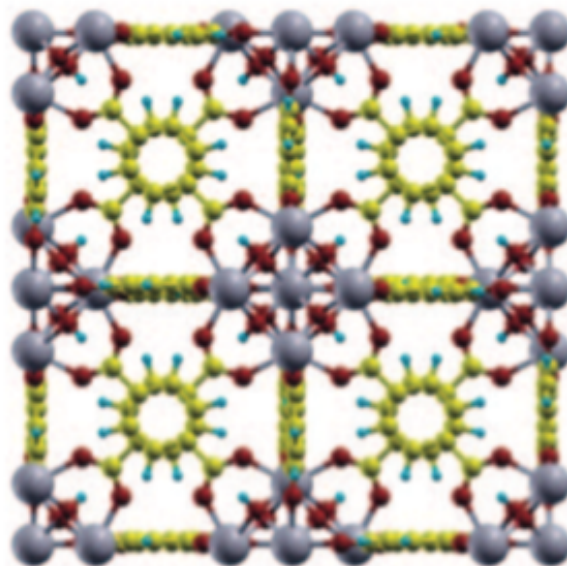


Exp. Spectrum: S. Bordiga and F. Bonino

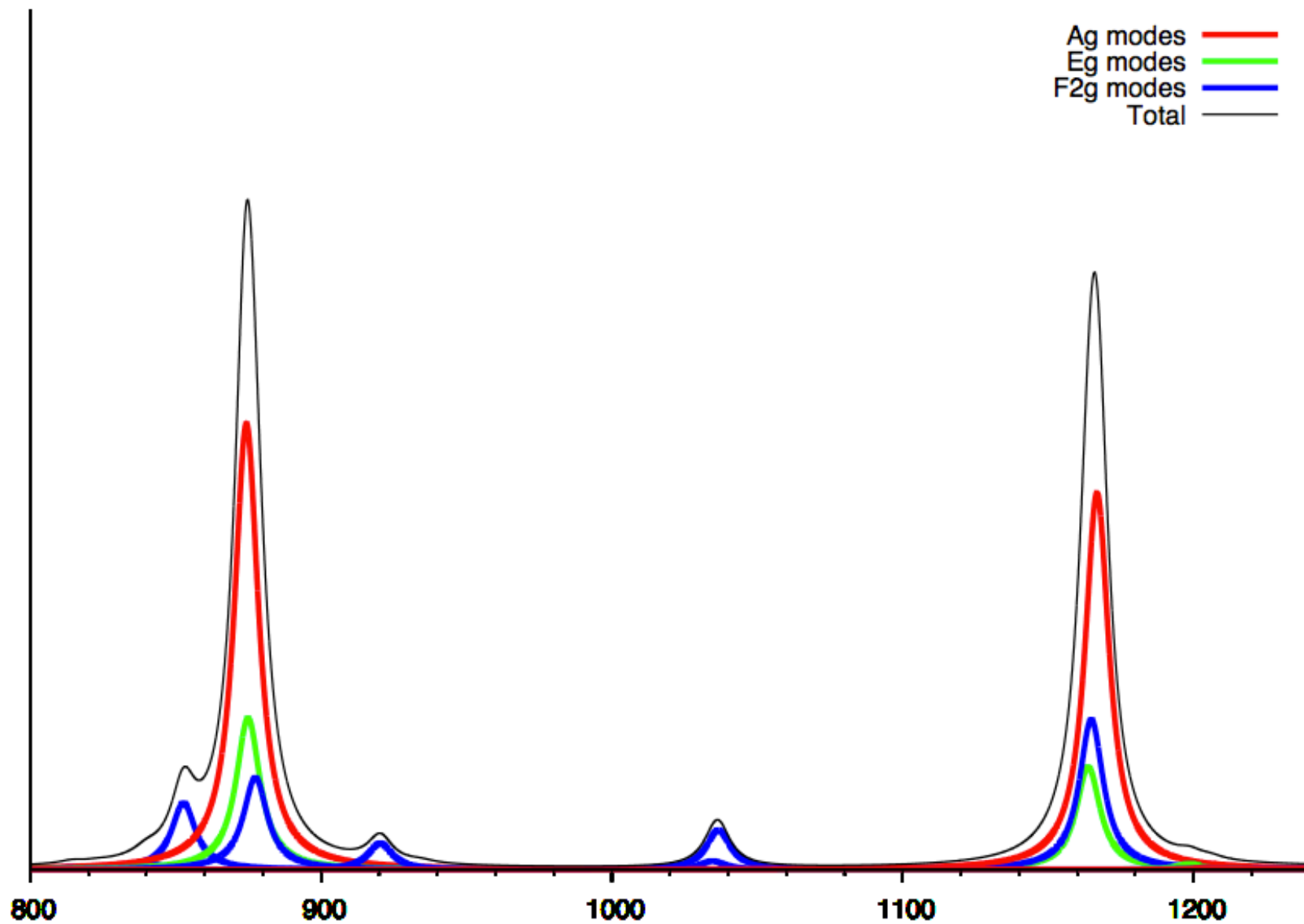
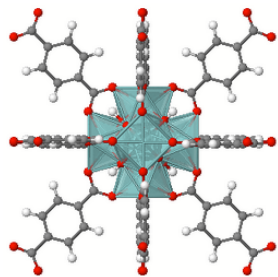
UiO-66 Metal Organic Framework

Experiment —
Calculated —

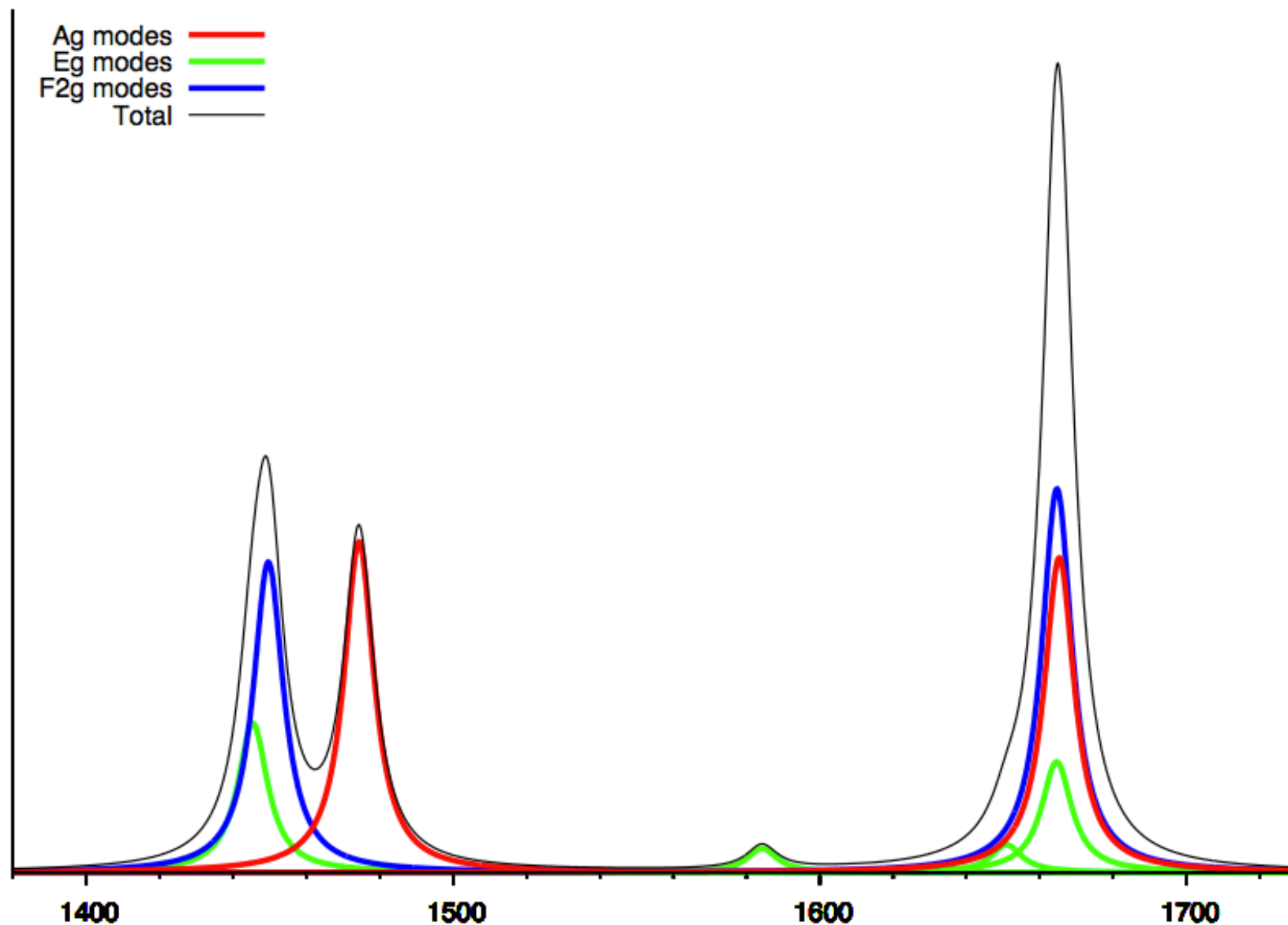
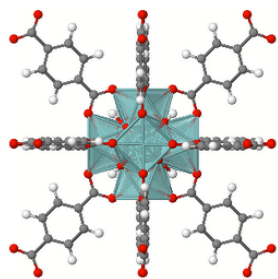
Better Synthesis



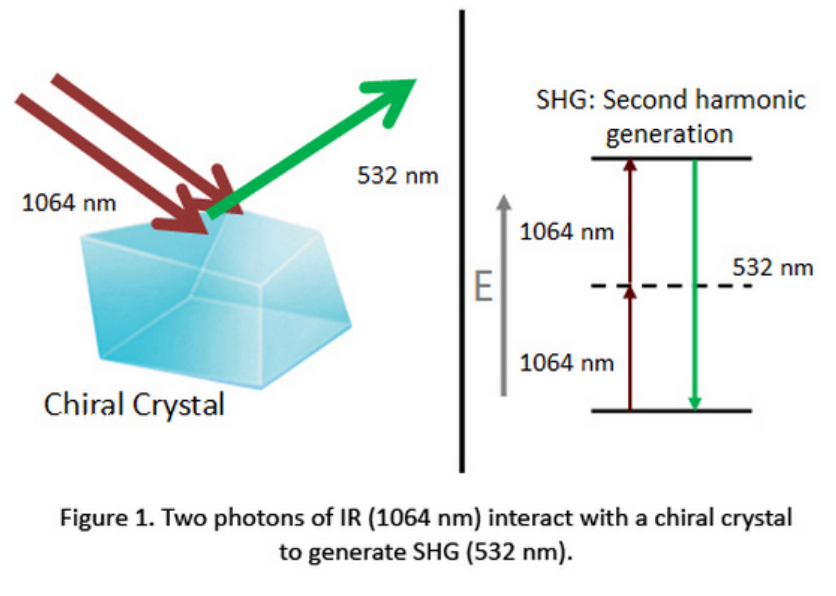
UiO-66 Metal Organic Framework



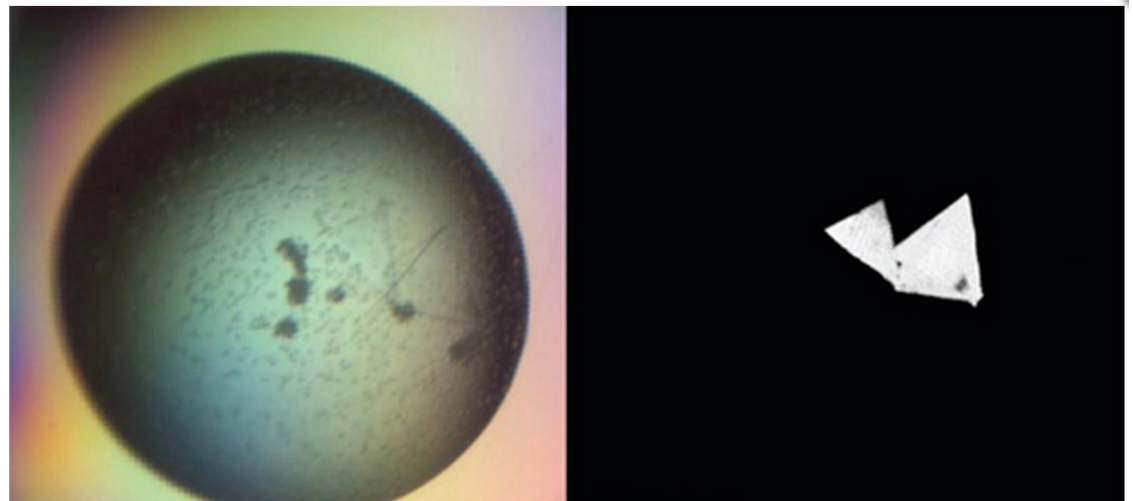
UiO-66 Metal Organic Framework



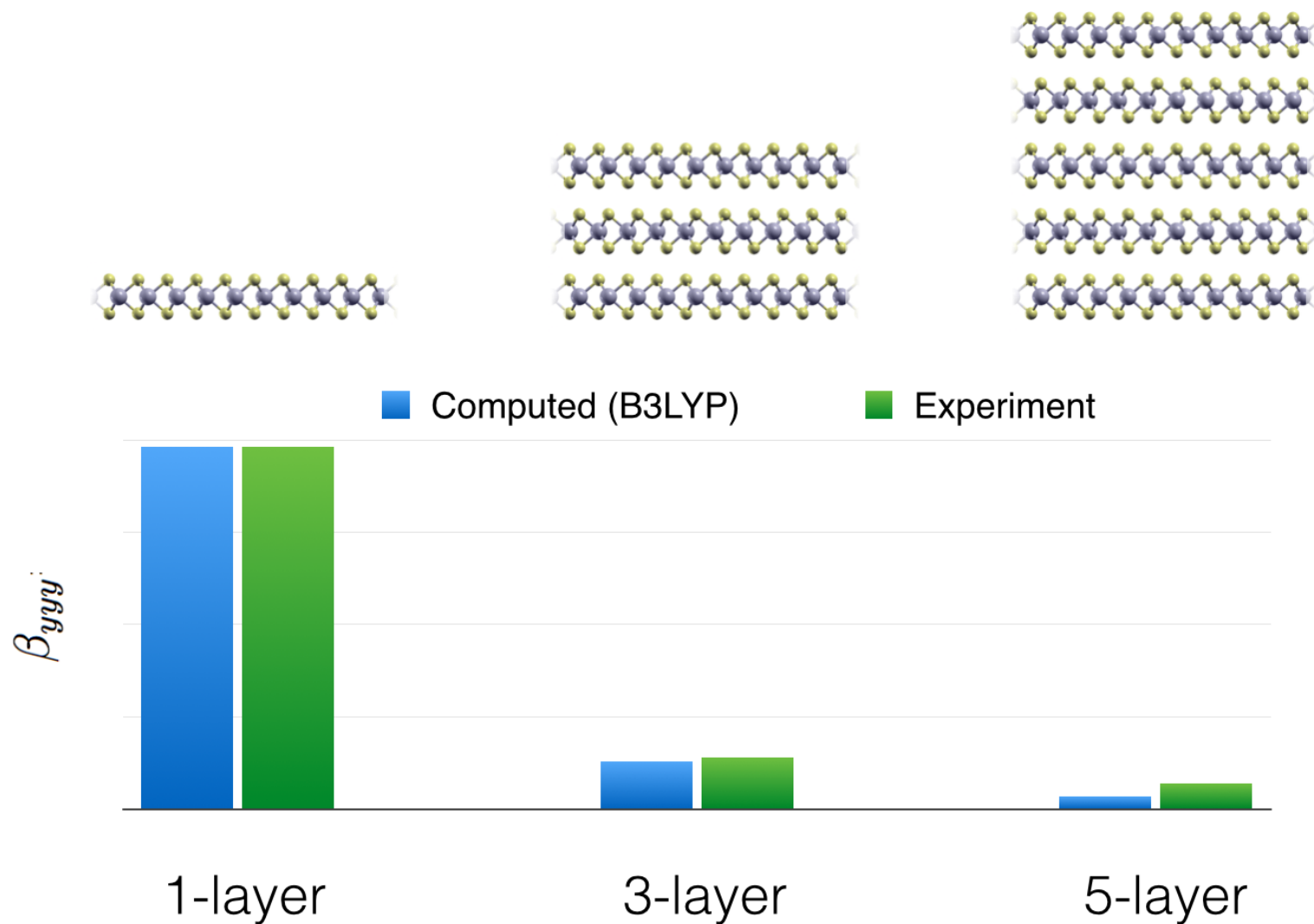
Second-Harmonic generation in molecular crystals



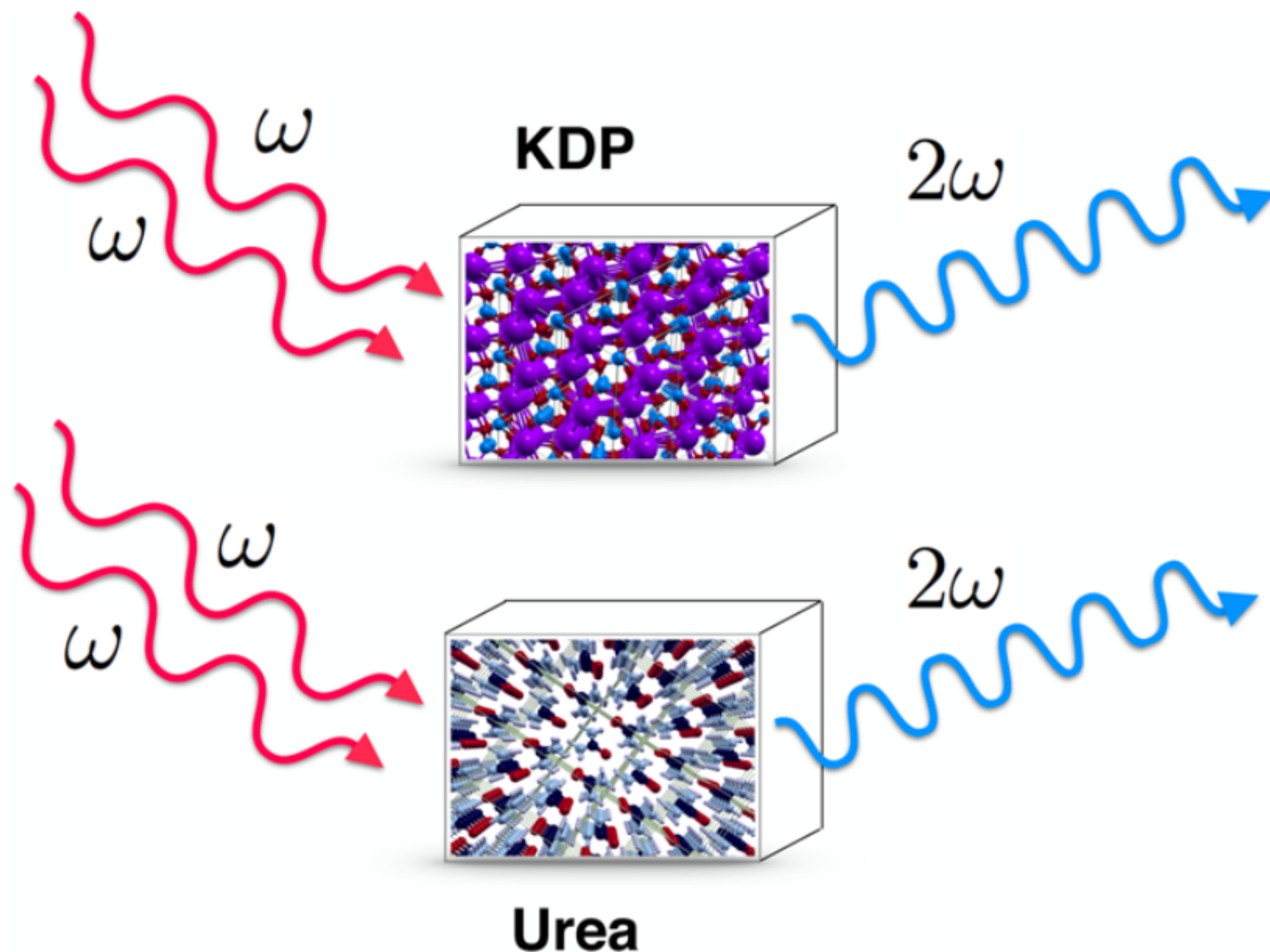
SHG imaging is a powerful experimental tool to find molecular crystals



Second Harmonic Generation: MoS₂ multi-layer



Second Harmonic Generation: Urea and KDP



Urea: SHG tensor components

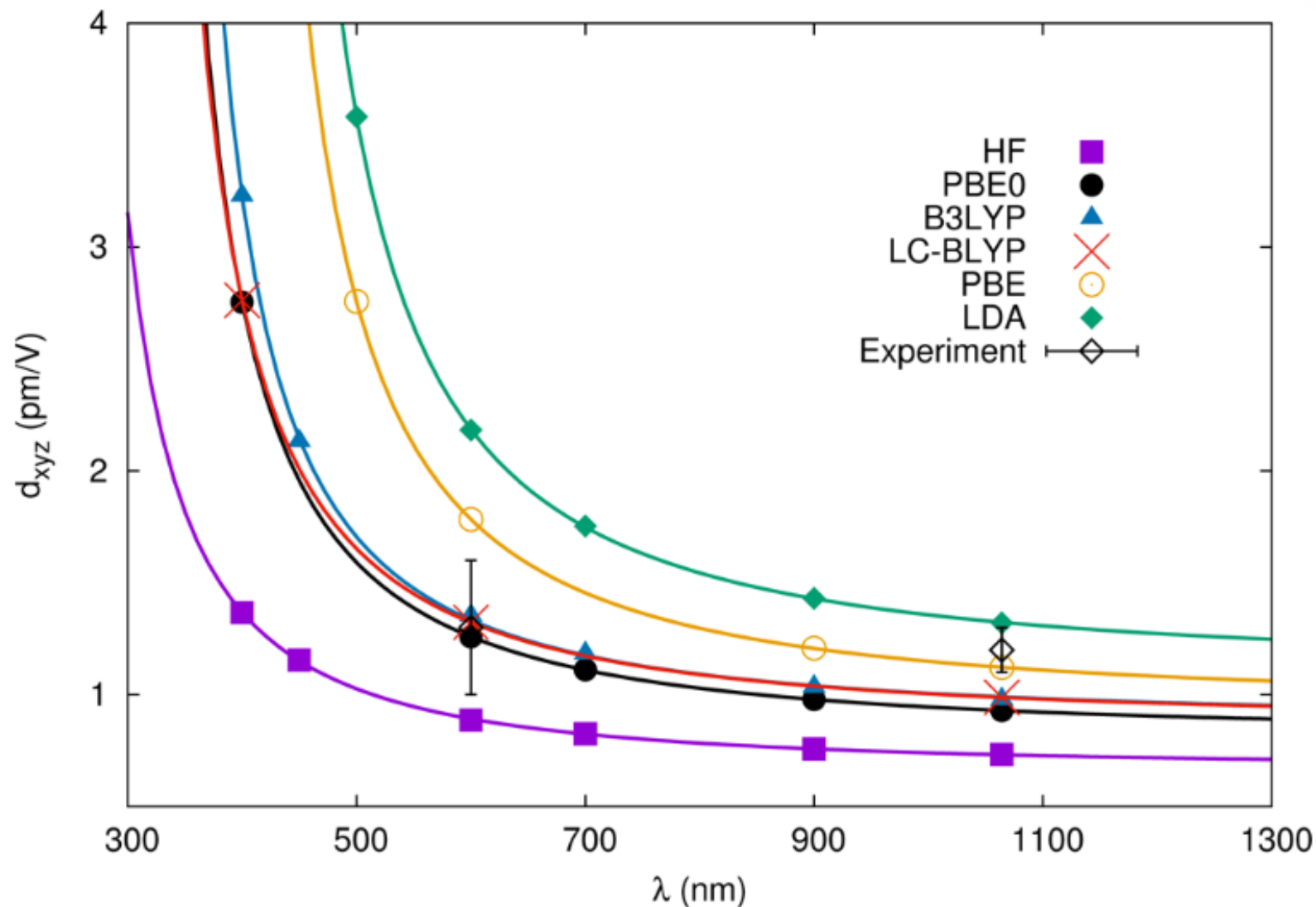


Figure 1. SHG d_{xyz} tensor component of bulk urea obtained with various functionals. The computed data are fit to the three parameter function: $d(\lambda) = a + b(1/(\omega_0 - 2\omega)(\omega_0 - \omega) + 1/(\omega_0 + \omega)(\omega_0 - \omega) + 1/(\omega_0 + \omega)(\omega_0 + 2\omega))$.

5. Coffee - Conclusions



Conclusions



Infrared and Raman spectra can be fully simulated with CRYSTAL. One input does it all, in a single run.



SHG and Pockels can be also simulated



Since all derivatives are performed analytically, the method is efficient and stable with respect to computational parameters



Comparison with experiments is very good

More Information

L. Maschio, B. Kirtman, R. Orlando, and M. R  rat “*Ab initio analytical infrared intensities for periodic systems through a coupled perturbed Hartree-Fock/Kohn-Sham method*” J. Chem. Phys. **137**, 204113 (2012)

L. Maschio, B. Kirtman, M. R  rat, R. Orlando, and R. Dovesi “*Ab initio analytical Raman intensities for periodic systems through a coupled perturbed Hartree-Fock/Kohn-Sham method I: theory*.” J. Chem. Phys. **139**, 164101 (2013)

L. Maschio, B. Kirtman, M. R  rat, R. Orlando, and R. Dovesi “*Ab initio analytical Raman intensities for periodic systems through a coupled perturbed Hartree-Fock/Kohn-Sham method II: validation and comparison with experiments*.” J. Chem. Phys. **139**, 164102 (2013)

L. Maschio, M. R  rat, B. Kirtman and R. Dovesi “Calculation of the dynamic first electronic hyperpolarizability $\beta(-\omega\sigma; \omega_1, \omega_2)$ of periodic systems. Theory, validation, and application to multi-layer MoS₂”, (2015) J. Chem. Phys. **143**, 244102

M. R  rat, L. Maschio, B. Kirtman, B. Civalleri, and R. Dovesi “Computation of Second Harmonic Generation for Crystalline Urea and KDP. An ab Initio Approach through the Coupled Perturbed Hartree-Fock/Kohn-Sham Scheme” J. Chem. Theory Comput. (2016) **12** (1), 107-113

Thank you all for your attention!

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